

BFS Traversal

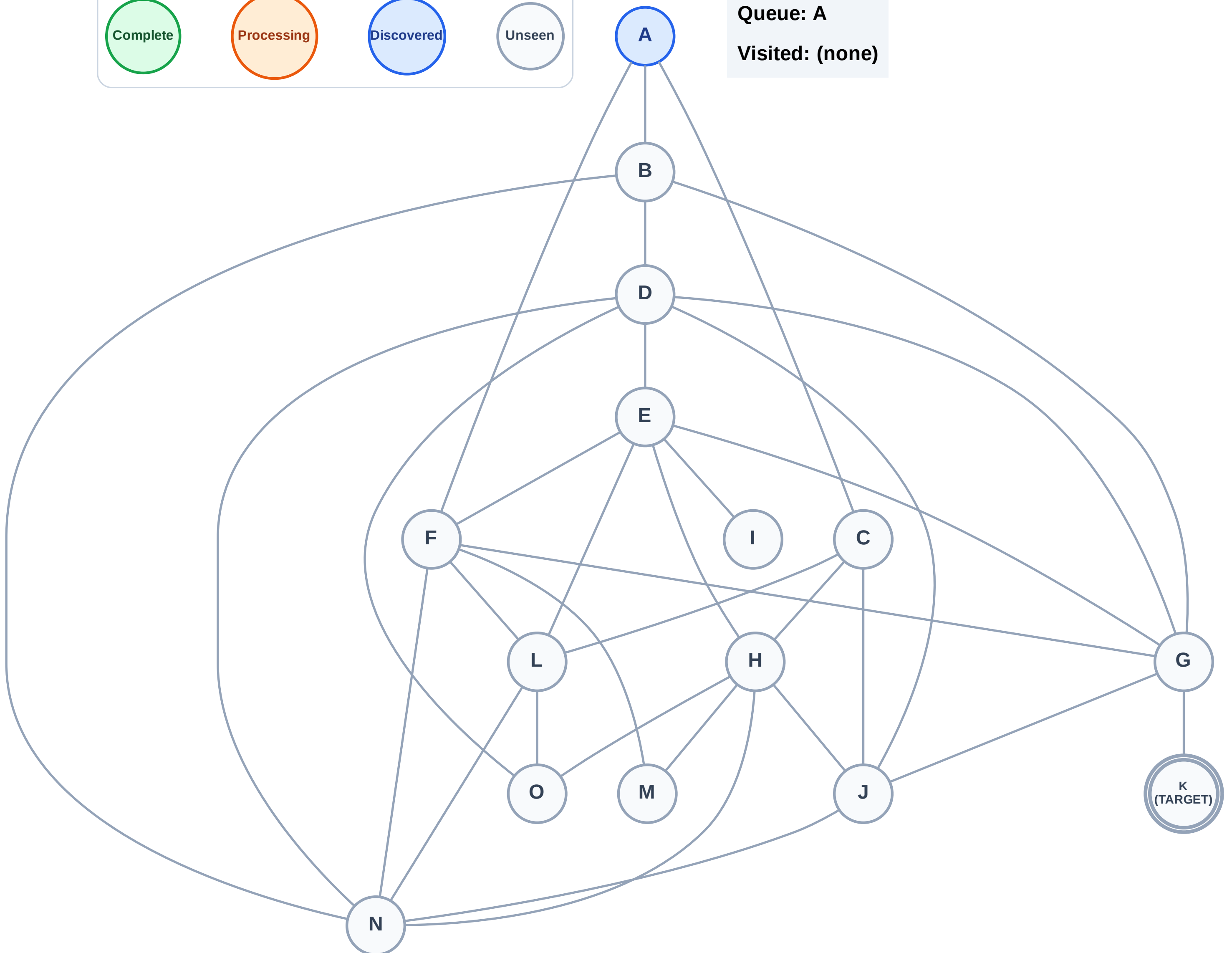
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



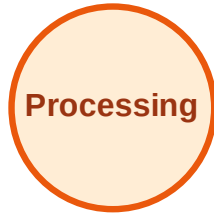
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



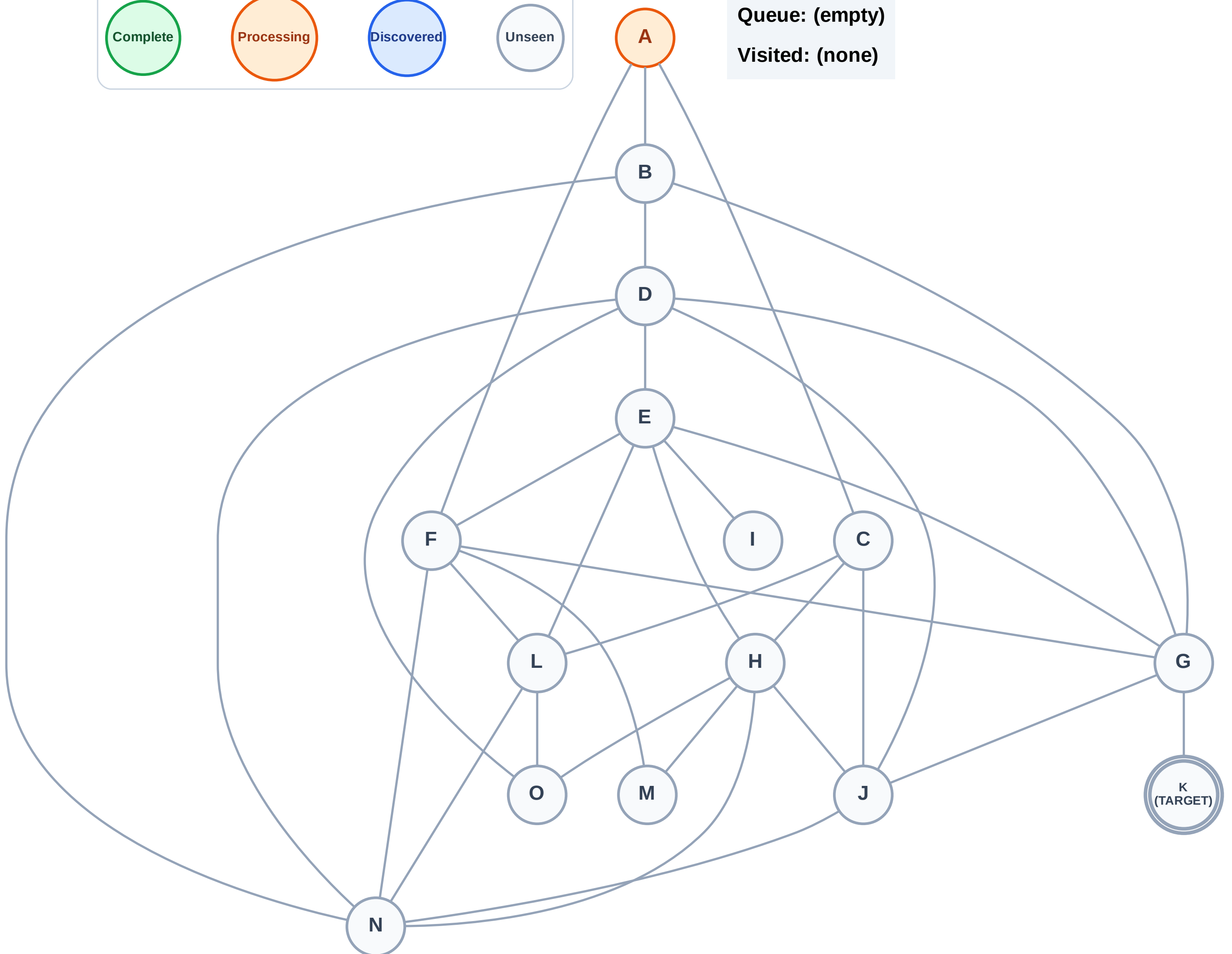
Discovered



Unseen

Queue: (empty)

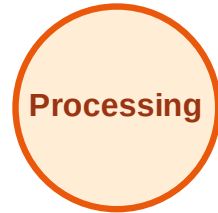
Visited: (none)



BFS Traversal

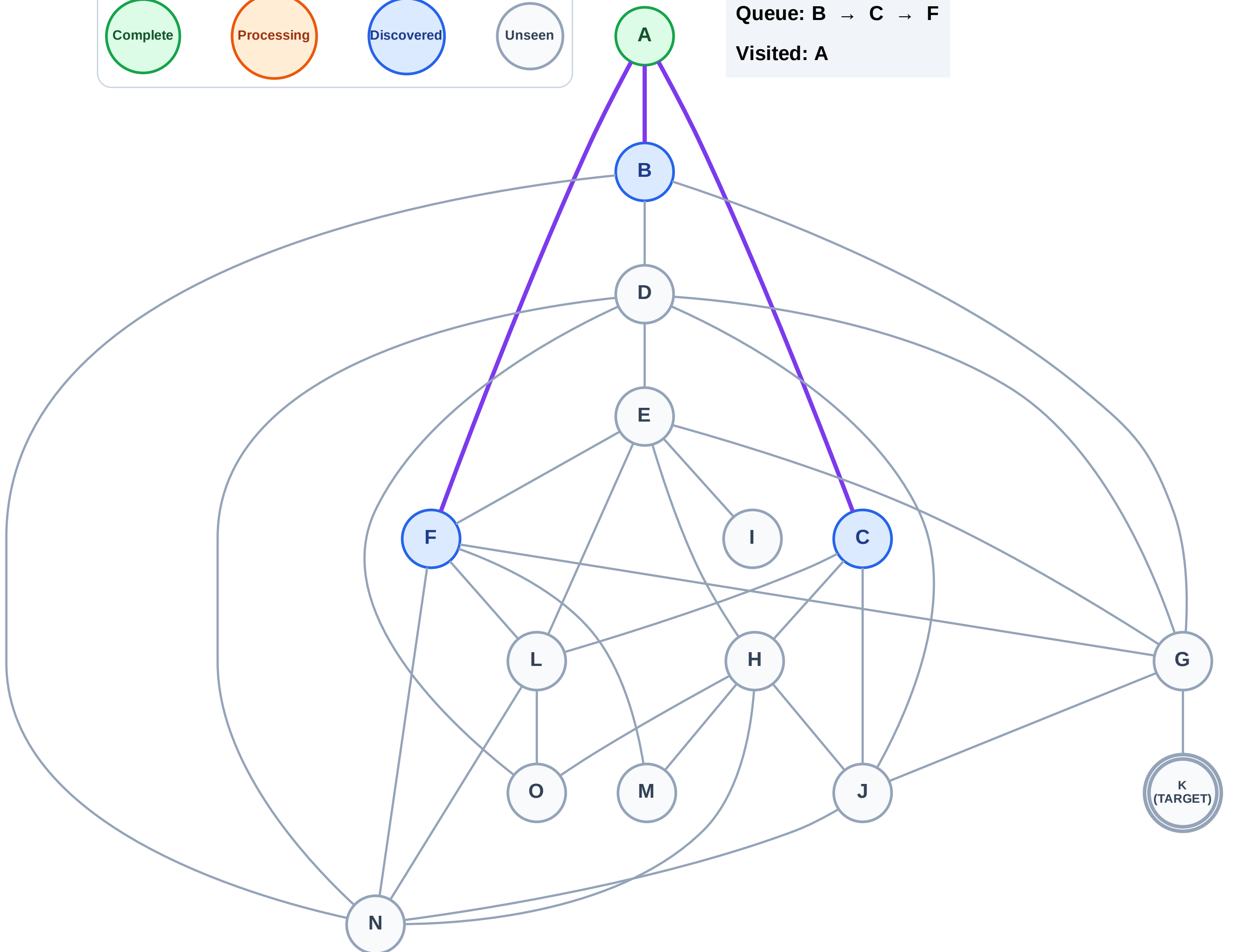
Step 3: Discover B, C, F from A

Legend



Queue: B → C → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

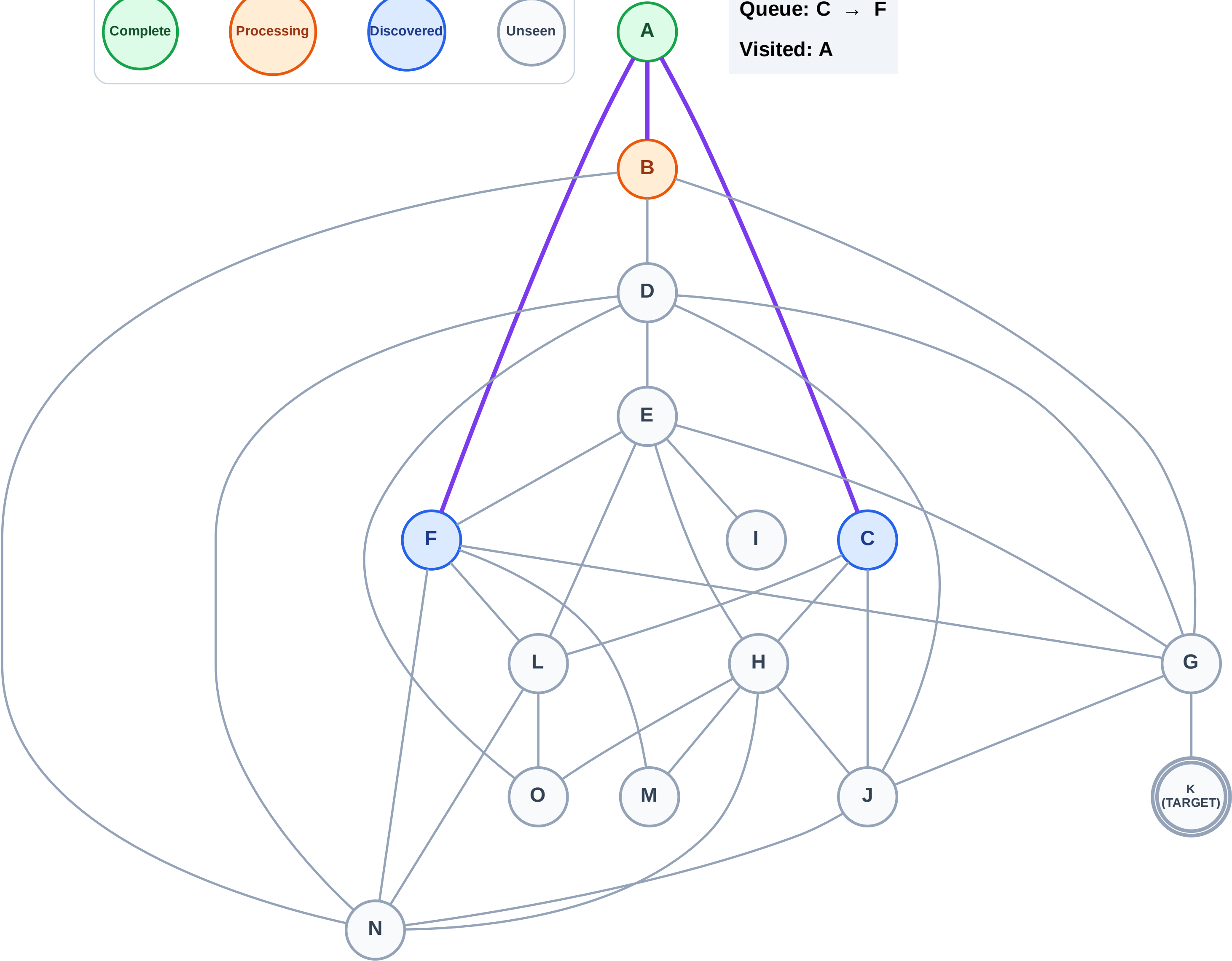
Processing

Discovered

Unseen

Queue: C → F

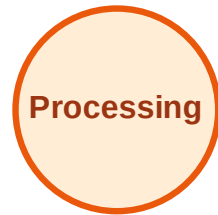
Visited: A



BFS Traversal

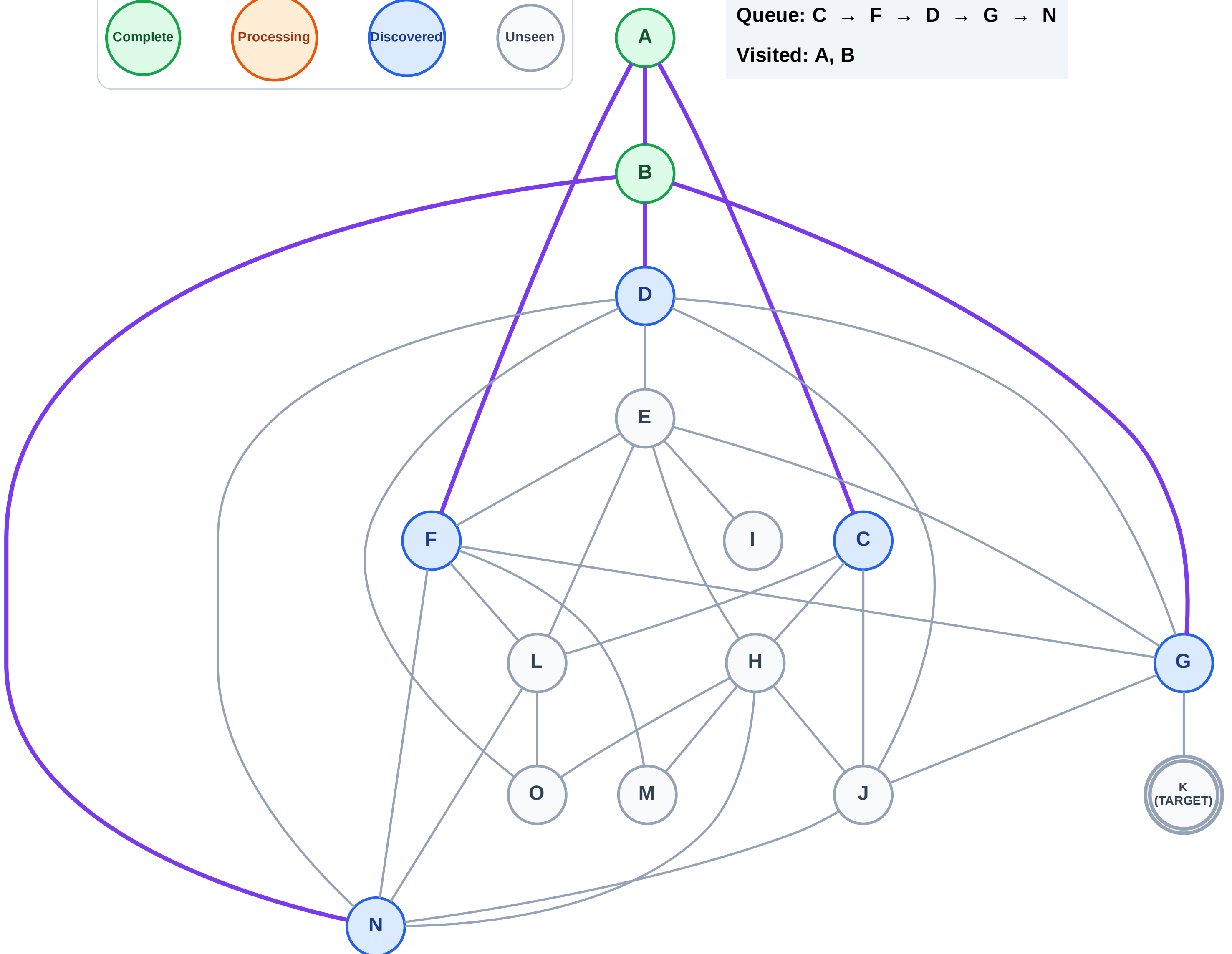
Step 5: Discover D, G, N from B

Legend



Queue: C → F → D → G → N

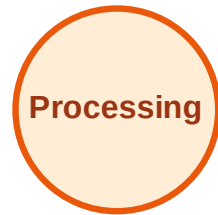
Visited: A, B



BFS Traversal

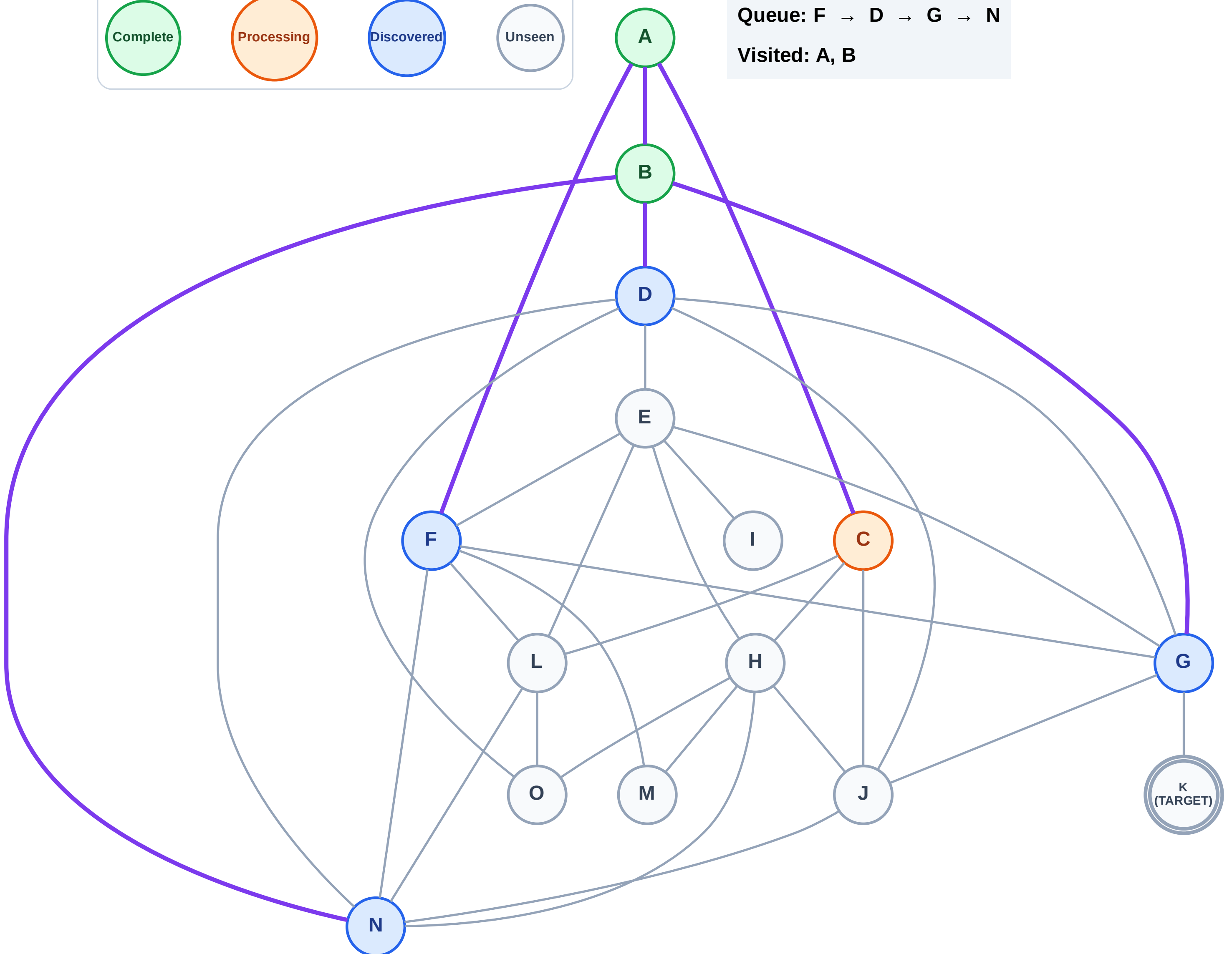
Step 6: Process node C

Legend



Queue: F → D → G → N

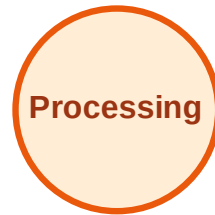
Visited: A, B



BFS Traversal

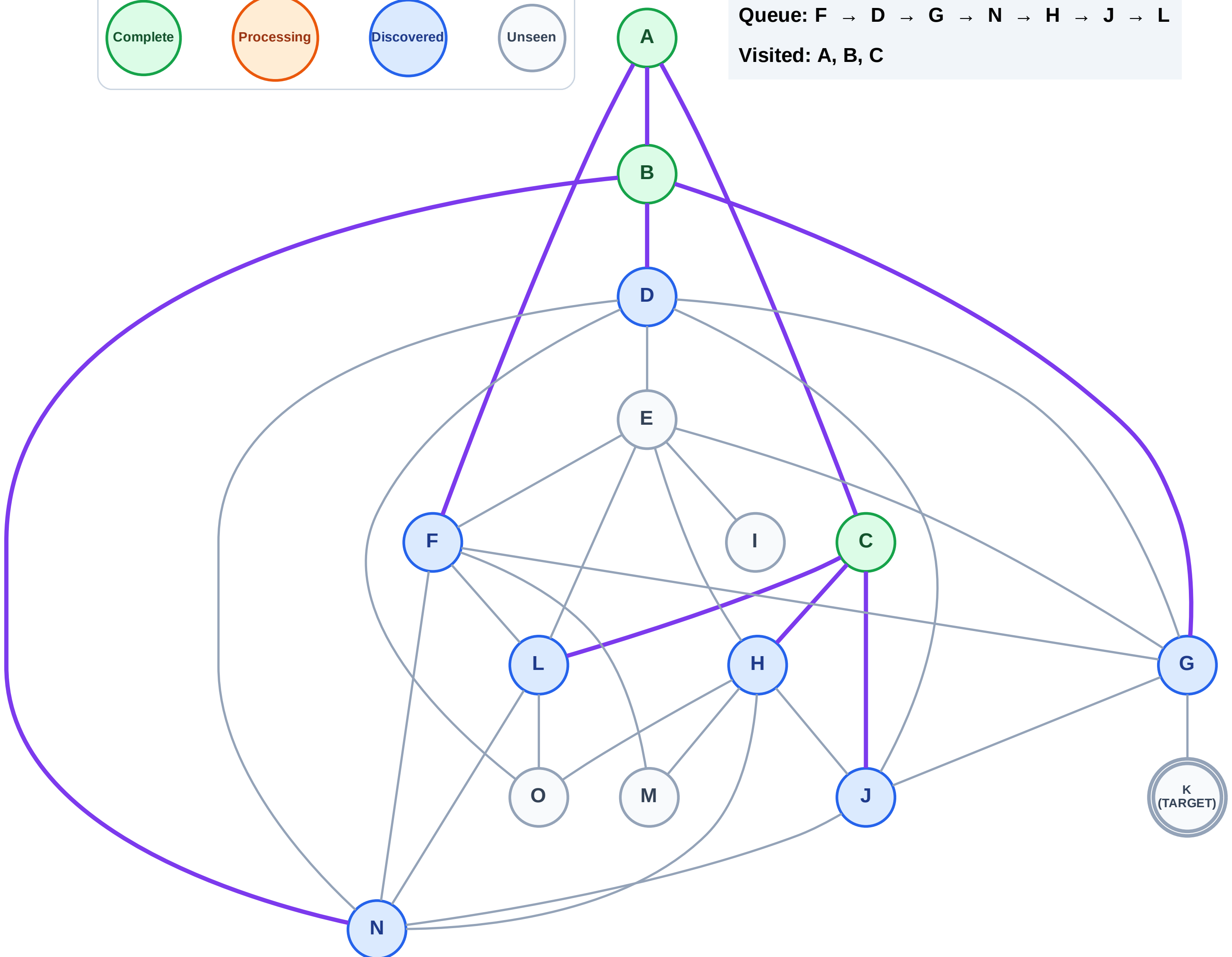
Step 7: Discover H, J, L from C

Legend



Queue: F → D → G → N → H → J → L

Visited: A, B, C



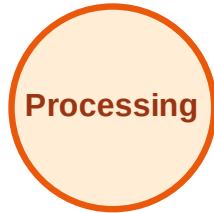
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



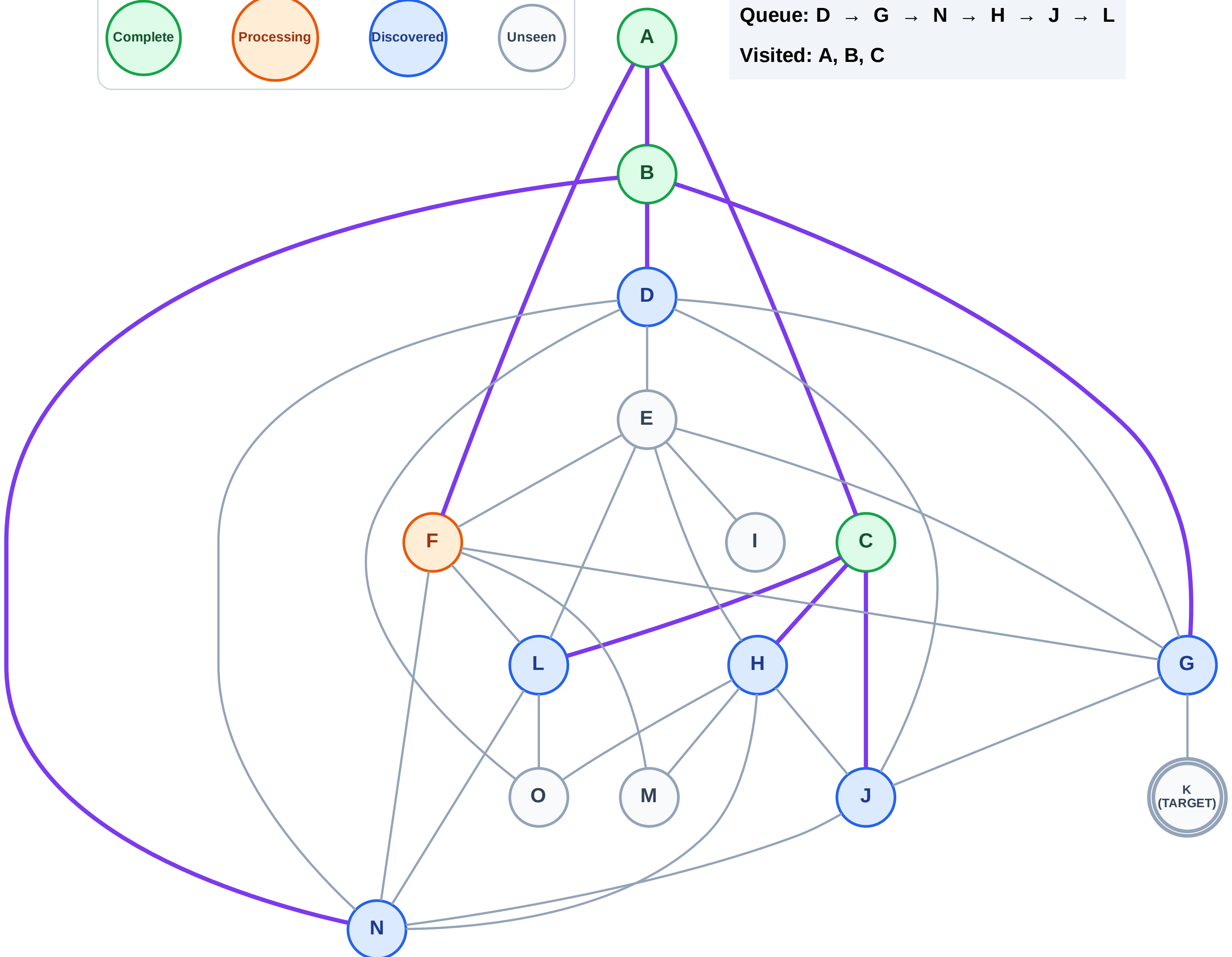
Discovered



Unseen

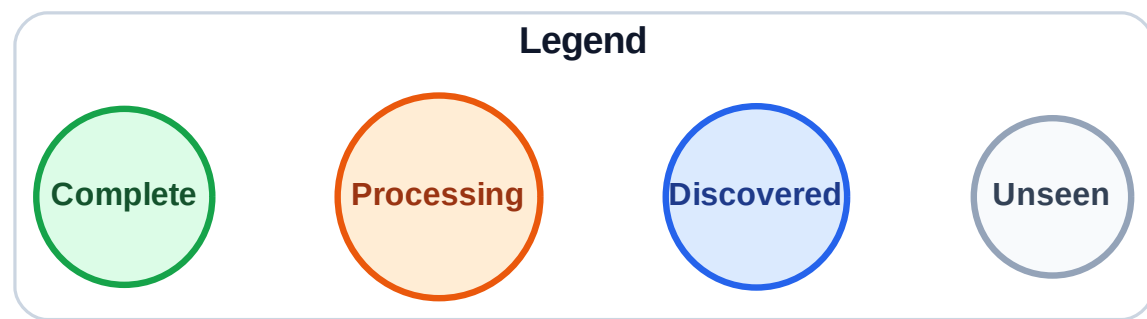
Queue: D → G → N → H → J → L

Visited: A, B, C



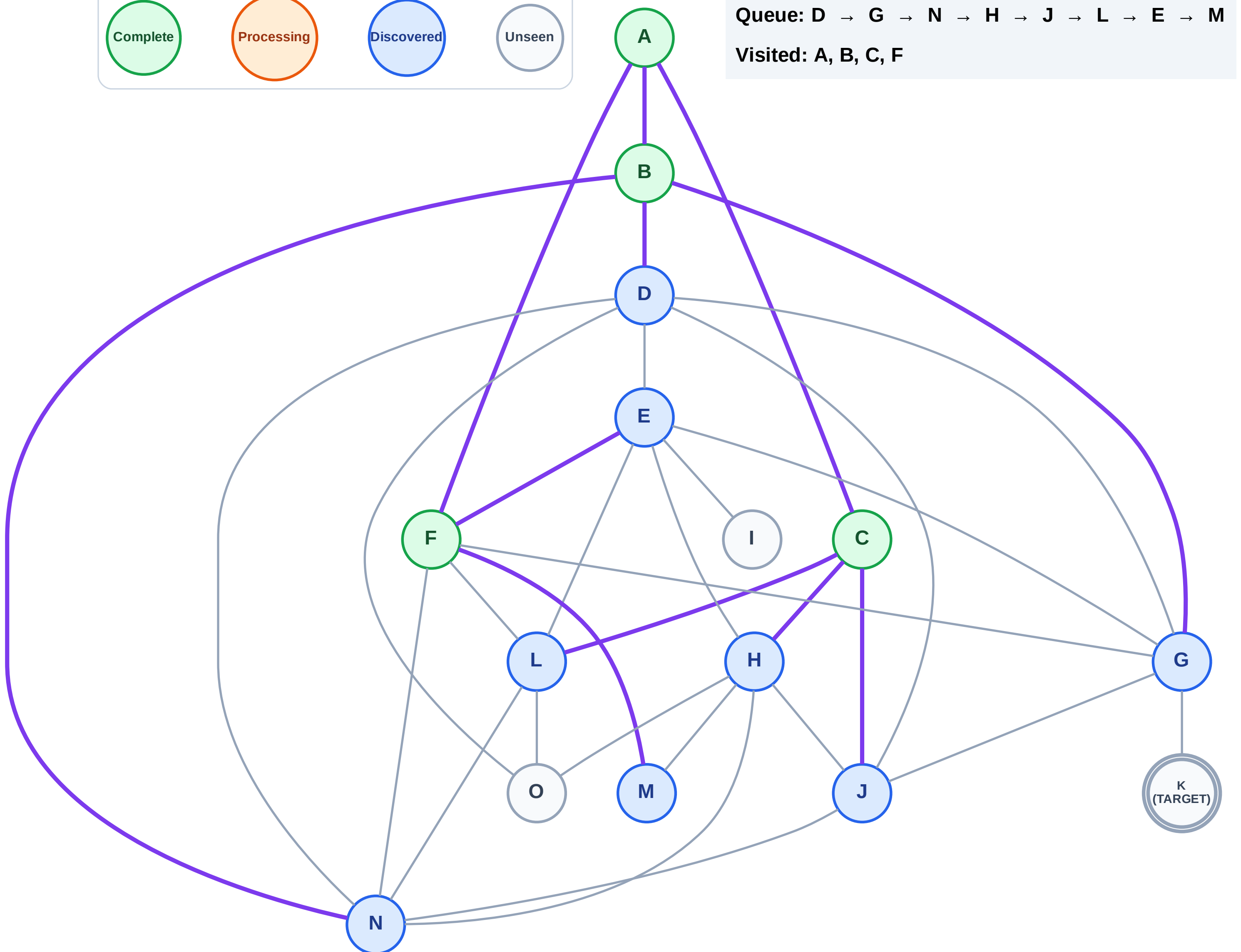
BFS Traversal

Step 9: Discover E, M from F



Queue: D → G → N → H → J → L → E → M

Visited: A, B, C, F

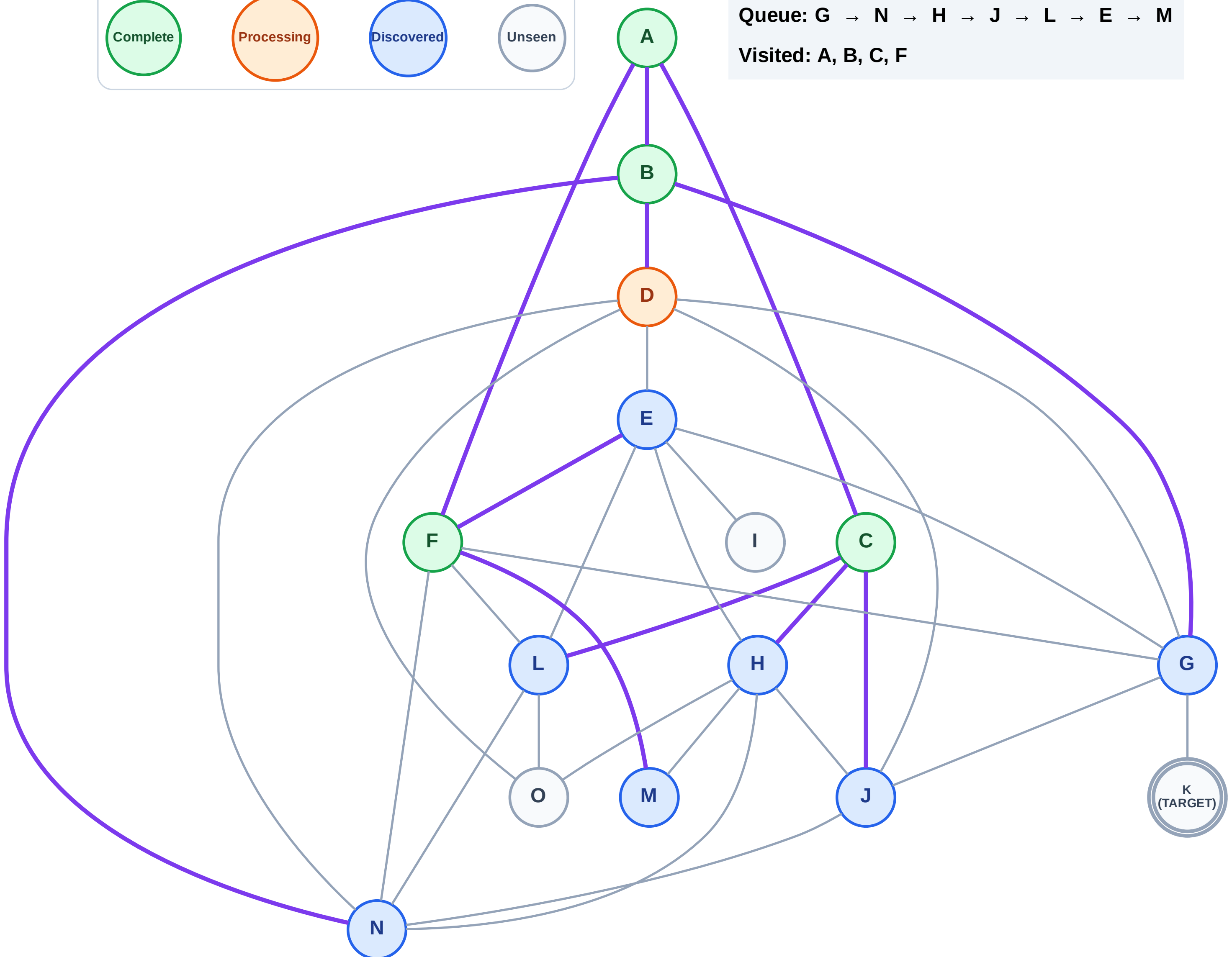


Step 10: Process node D

Legend

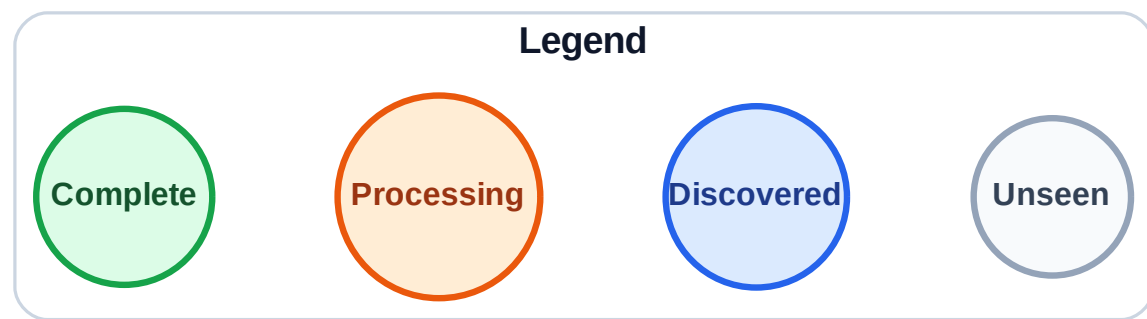
Unseen

Visited: A, B, C, F



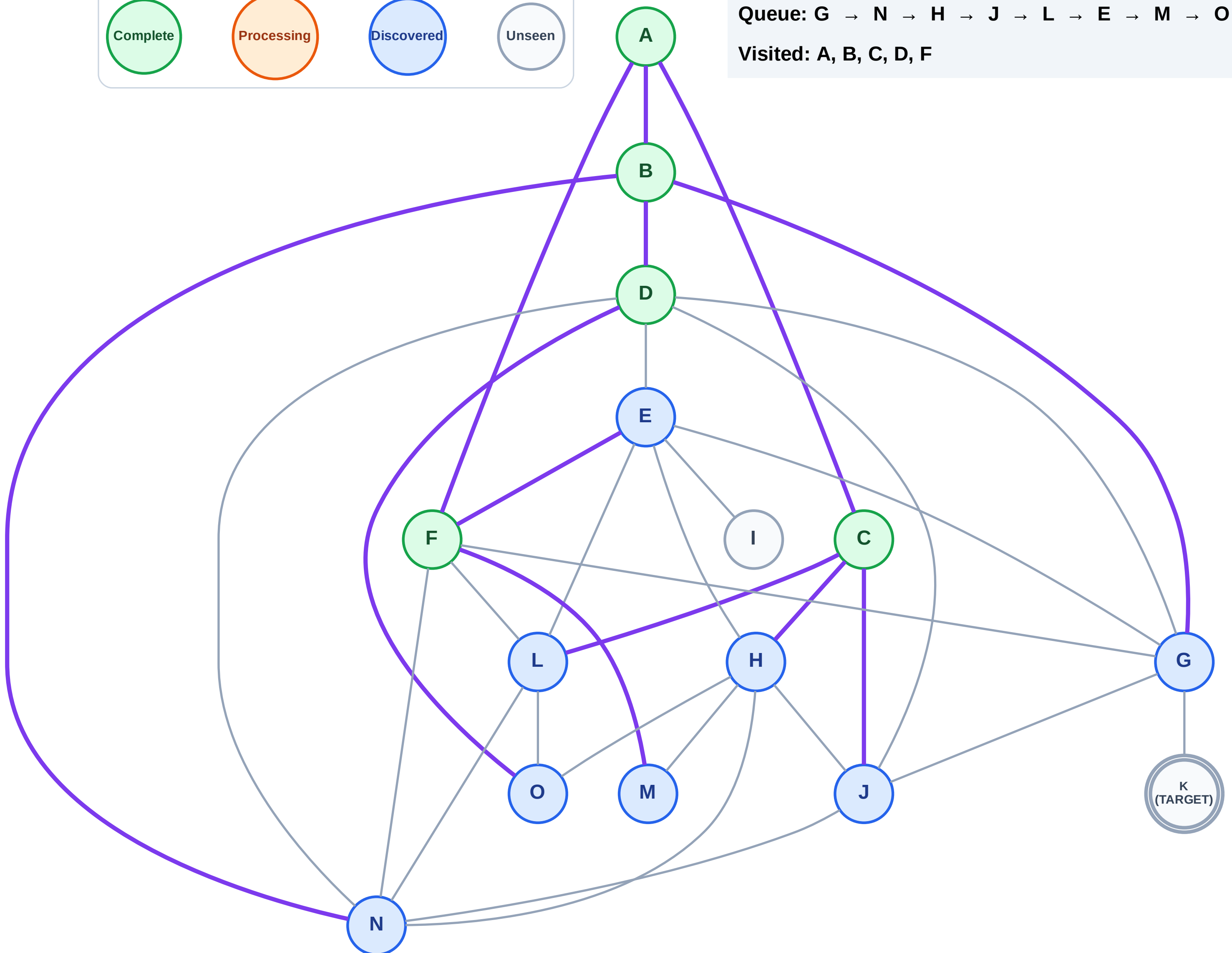
BFS Traversal

Step 11: Discover O from D

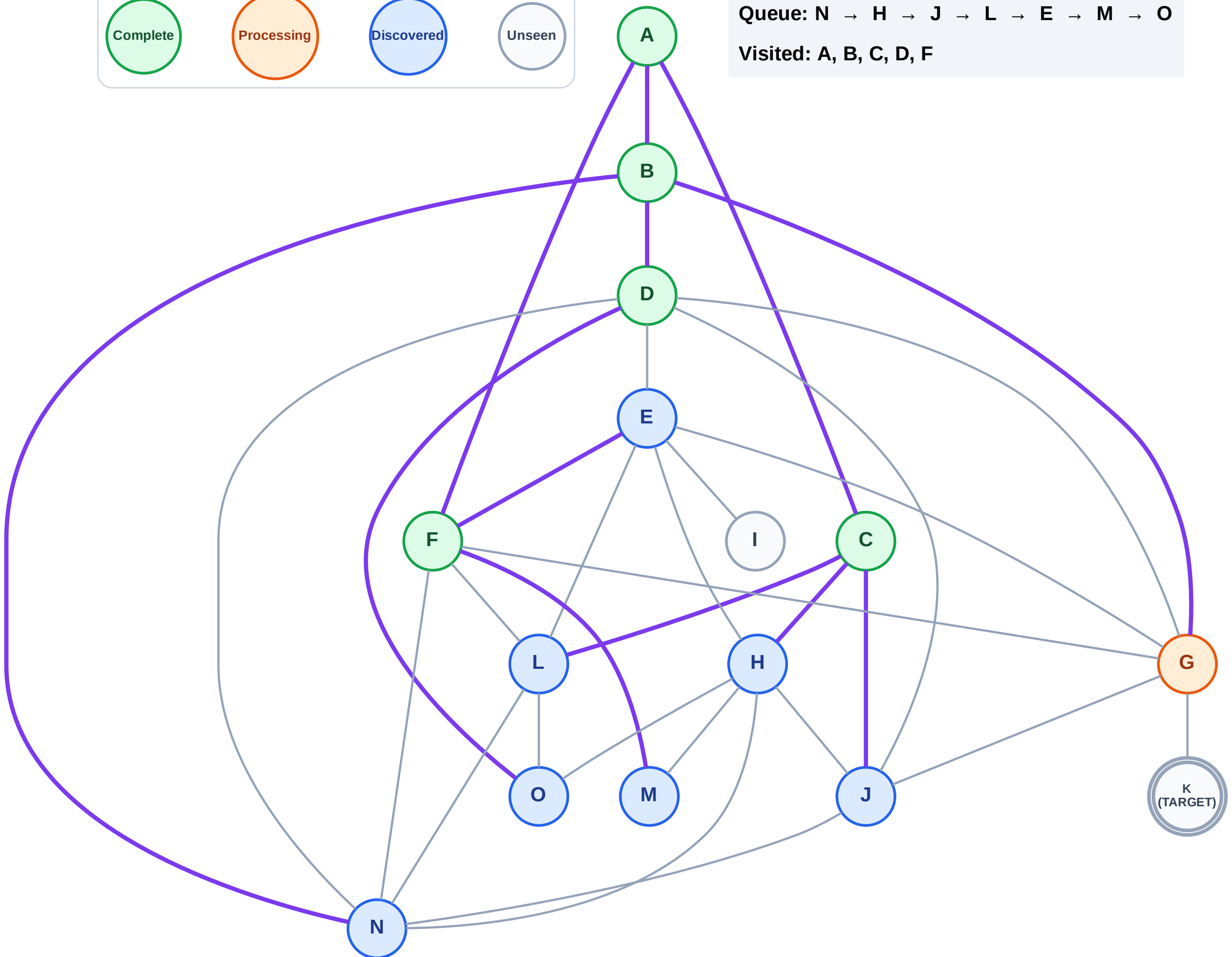


Queue: G → N → H → J → L → E → M → O

Visited: A, B, C, D, F



Step 12: Process node G



BFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → G → K

Legend

Complete

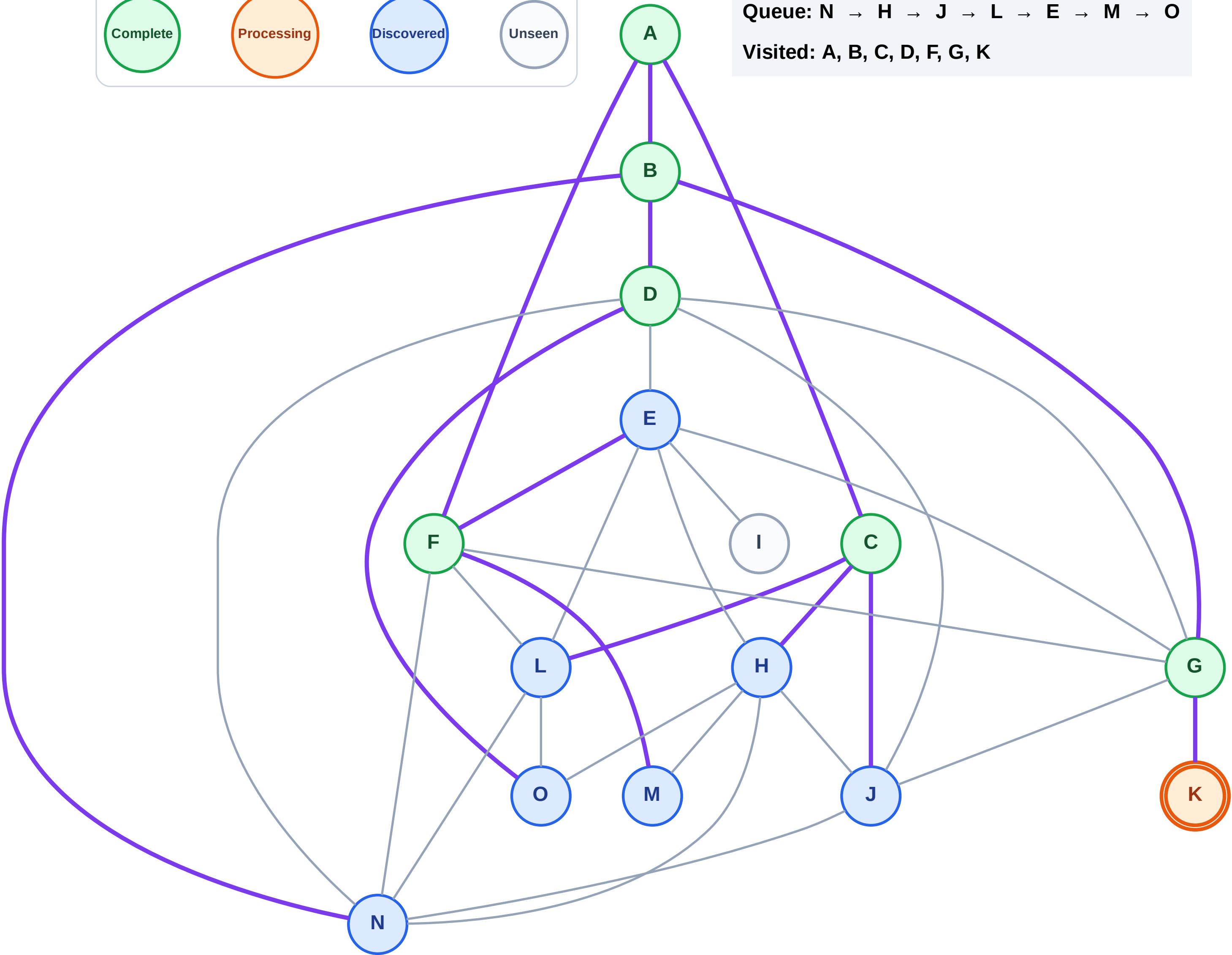
Processing

Discovered

Unseen

Queue: N → H → J → L → E → M → O

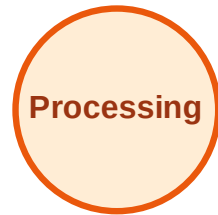
Visited: A, B, C, D, F, G, K



DFS Traversal

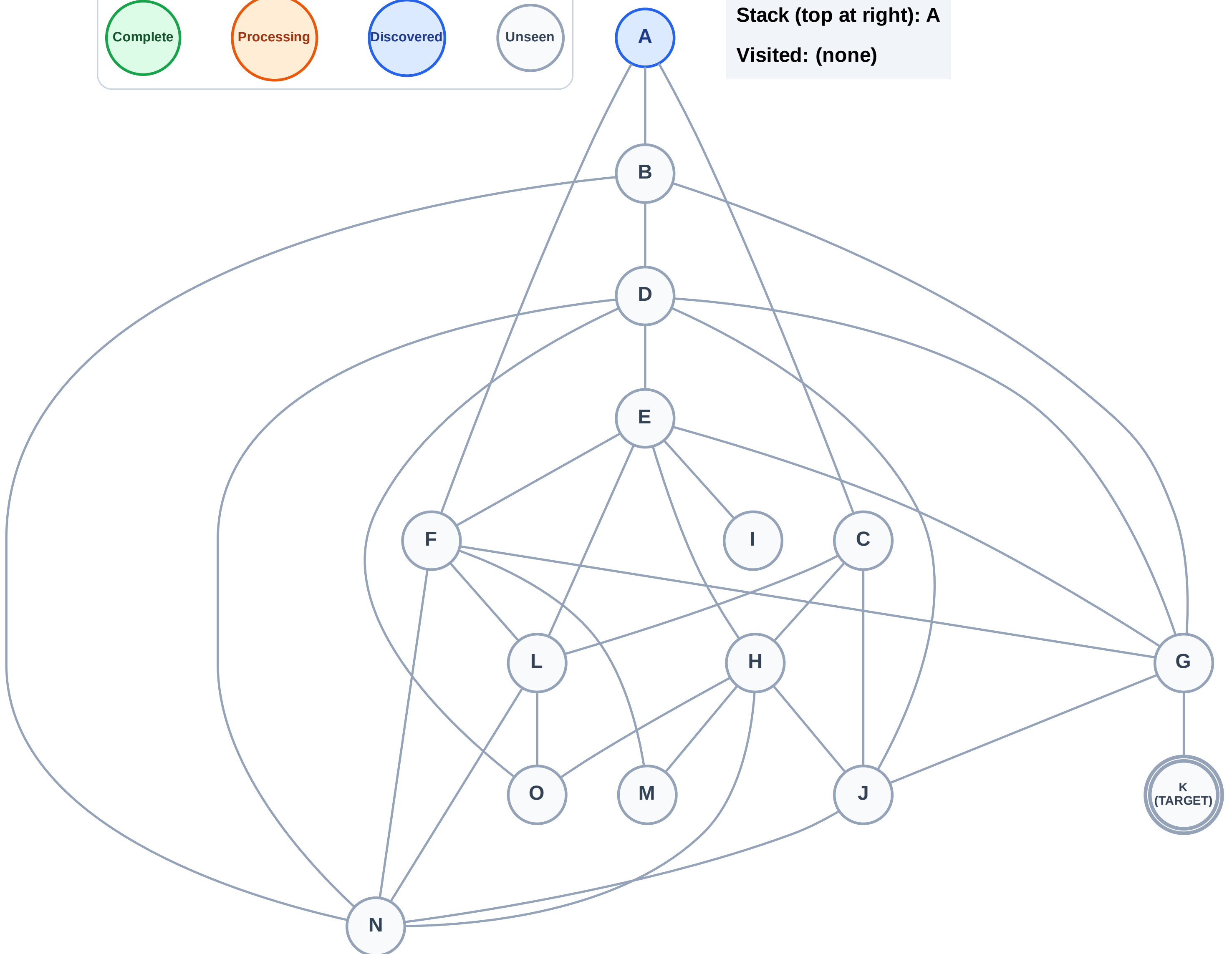
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

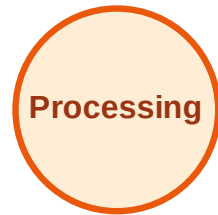
Visited: (none)



DFS Traversal

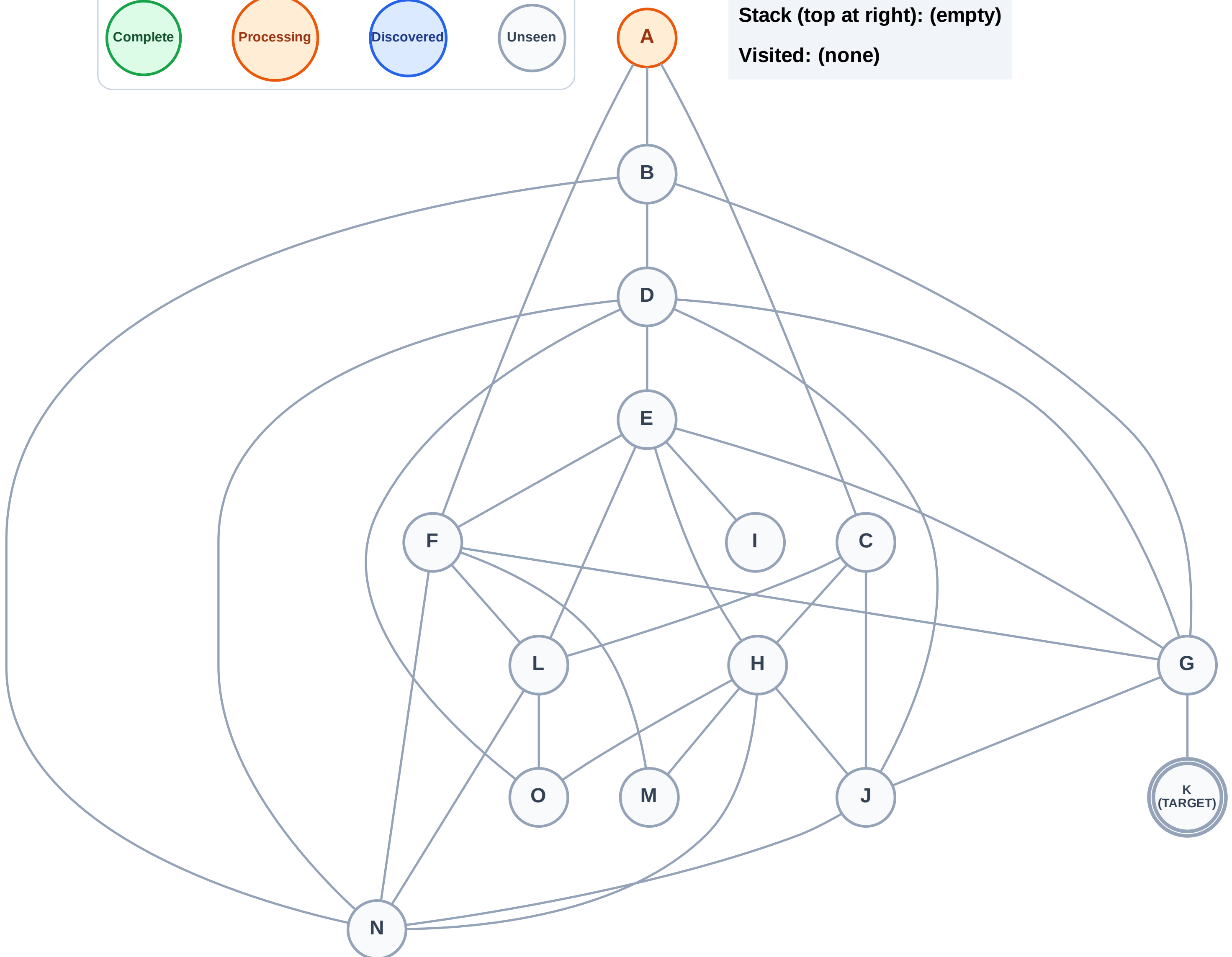
Step 2: Process node A

Legend



Stack (top at right): (empty)

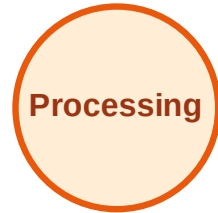
Visited: (none)



DFS Traversal

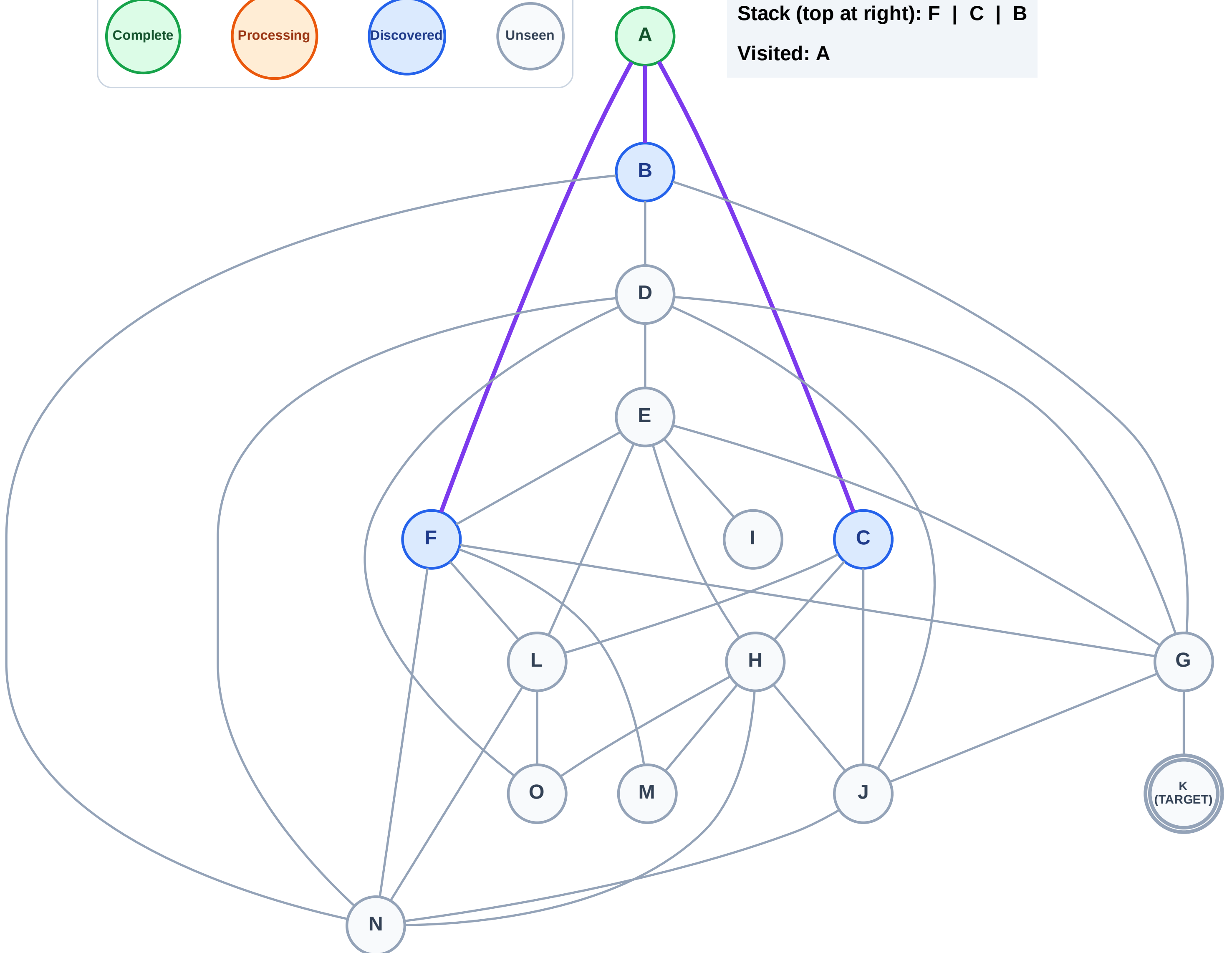
Step 3: Discover F, C, B from A

Legend



Stack (top at right): F | C | B

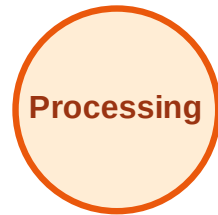
Visited: A



DFS Traversal

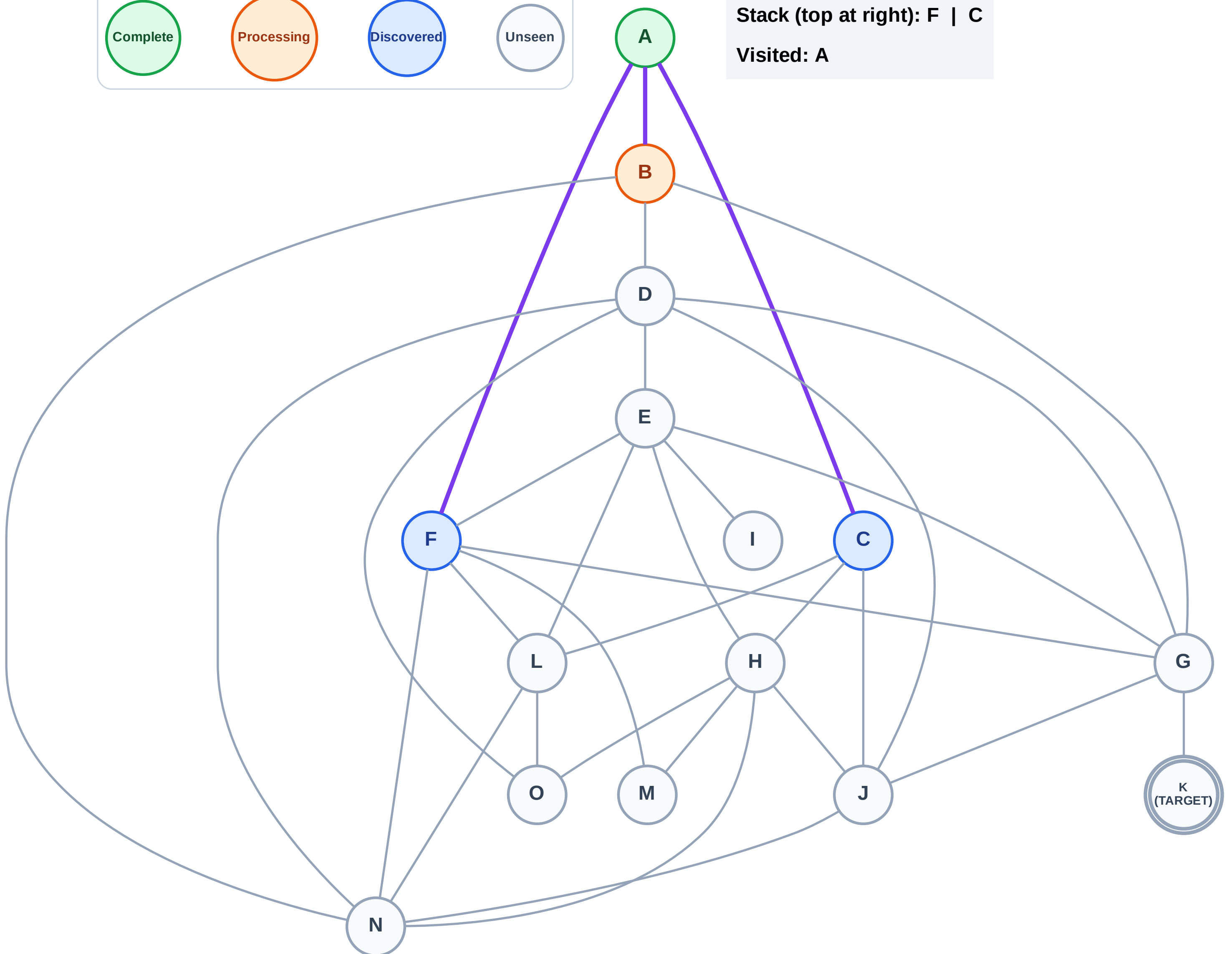
Step 4: Process node B

Legend



Stack (top at right): F | C

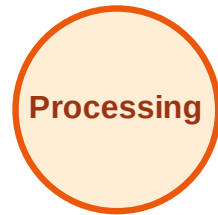
Visited: A



DFS Traversal

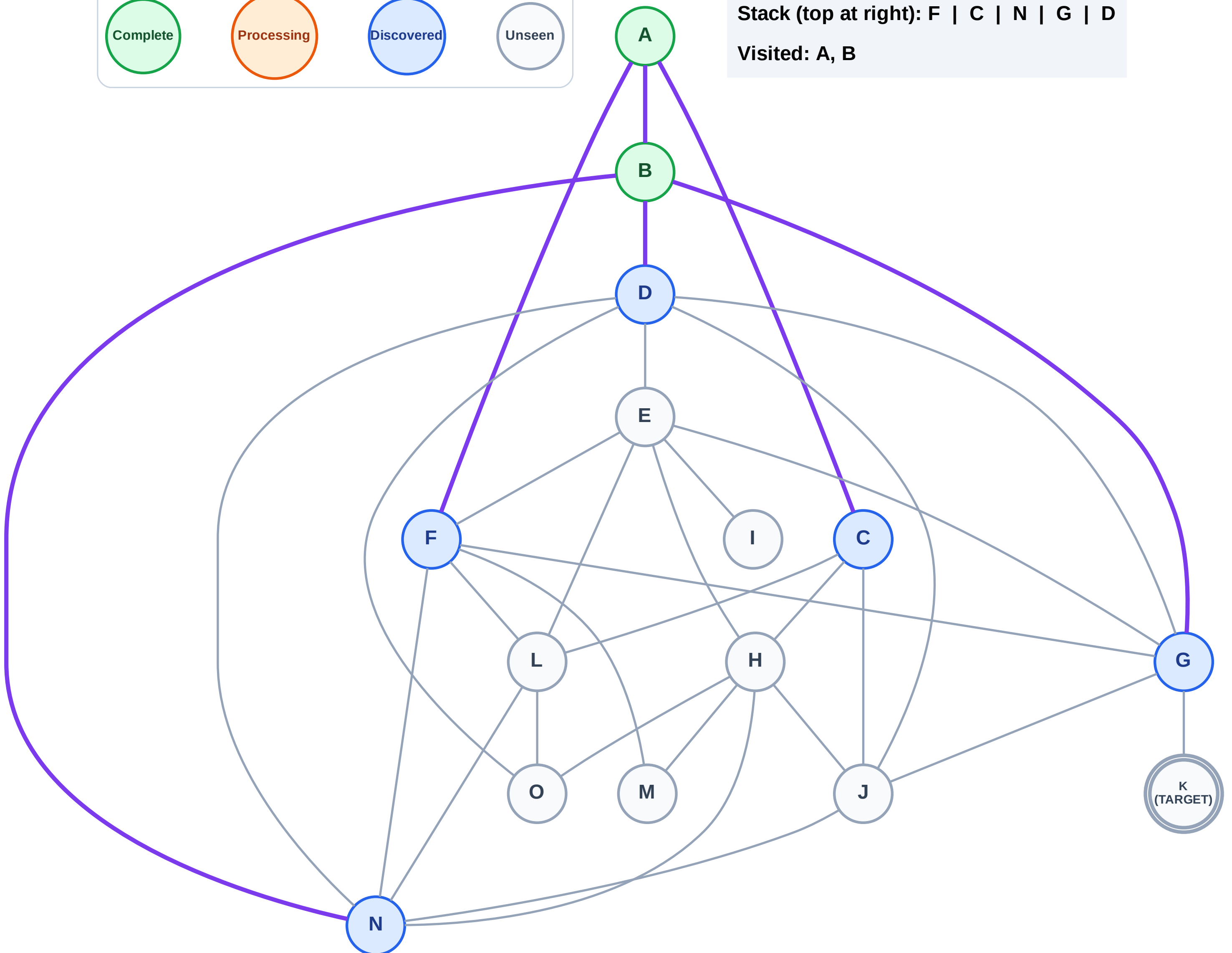
Step 5: Discover N, G, D from B

Legend



Stack (top at right): F | C | N | G | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

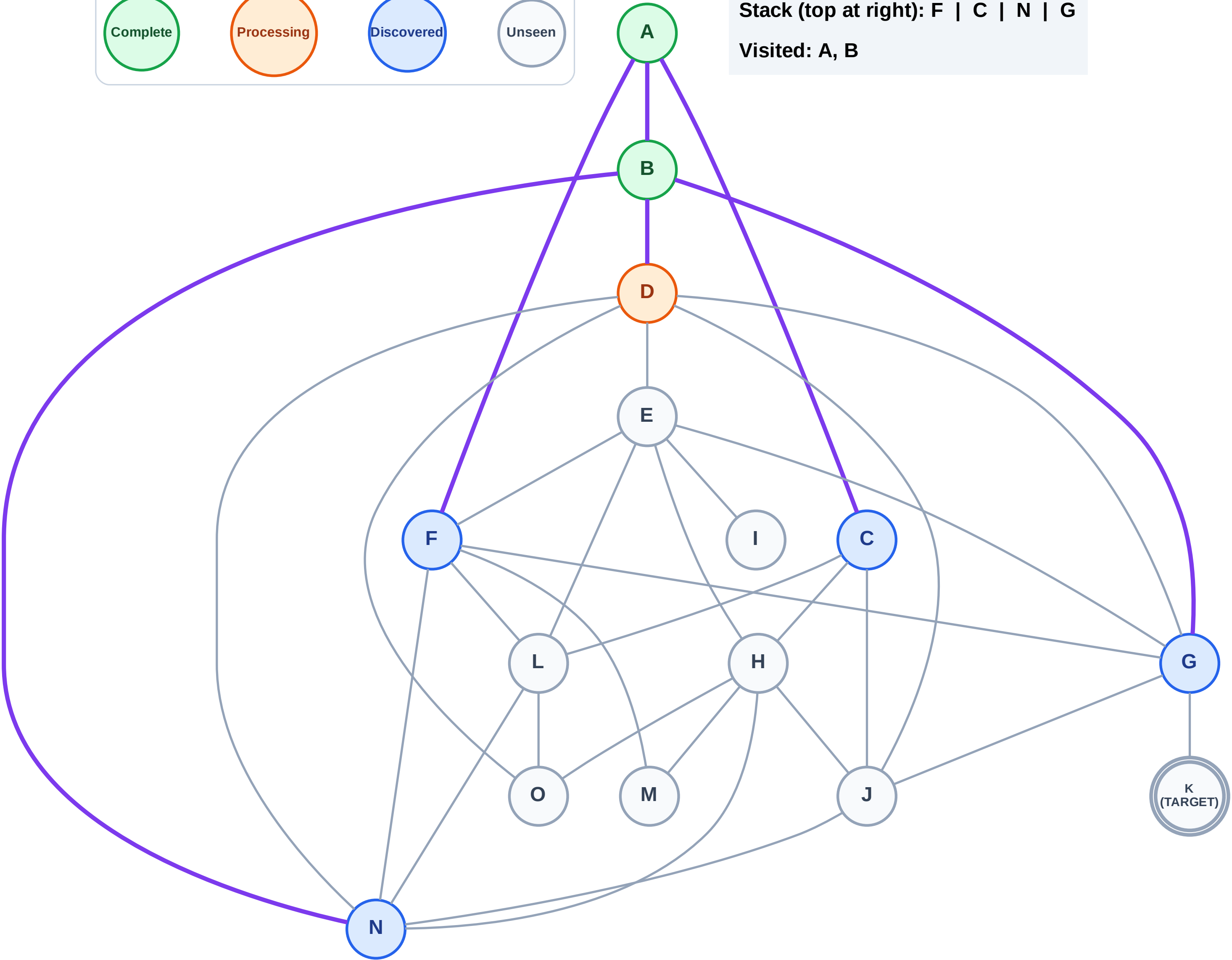
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G

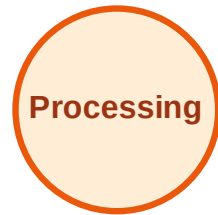
Visited: A, B



DFS Traversal

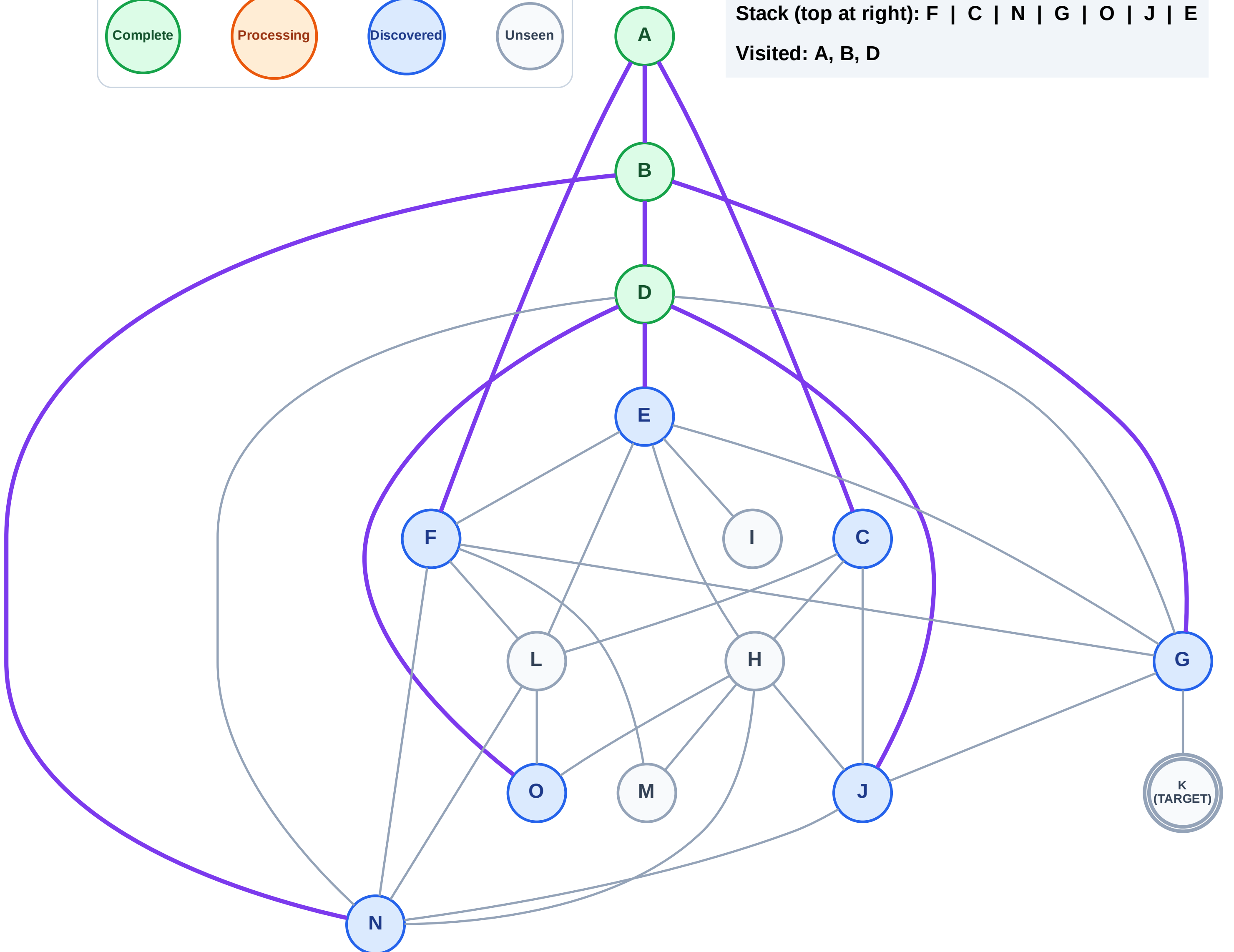
Step 7: Discover O, J, E from D

Legend



Stack (top at right): F | C | N | G | O | J | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

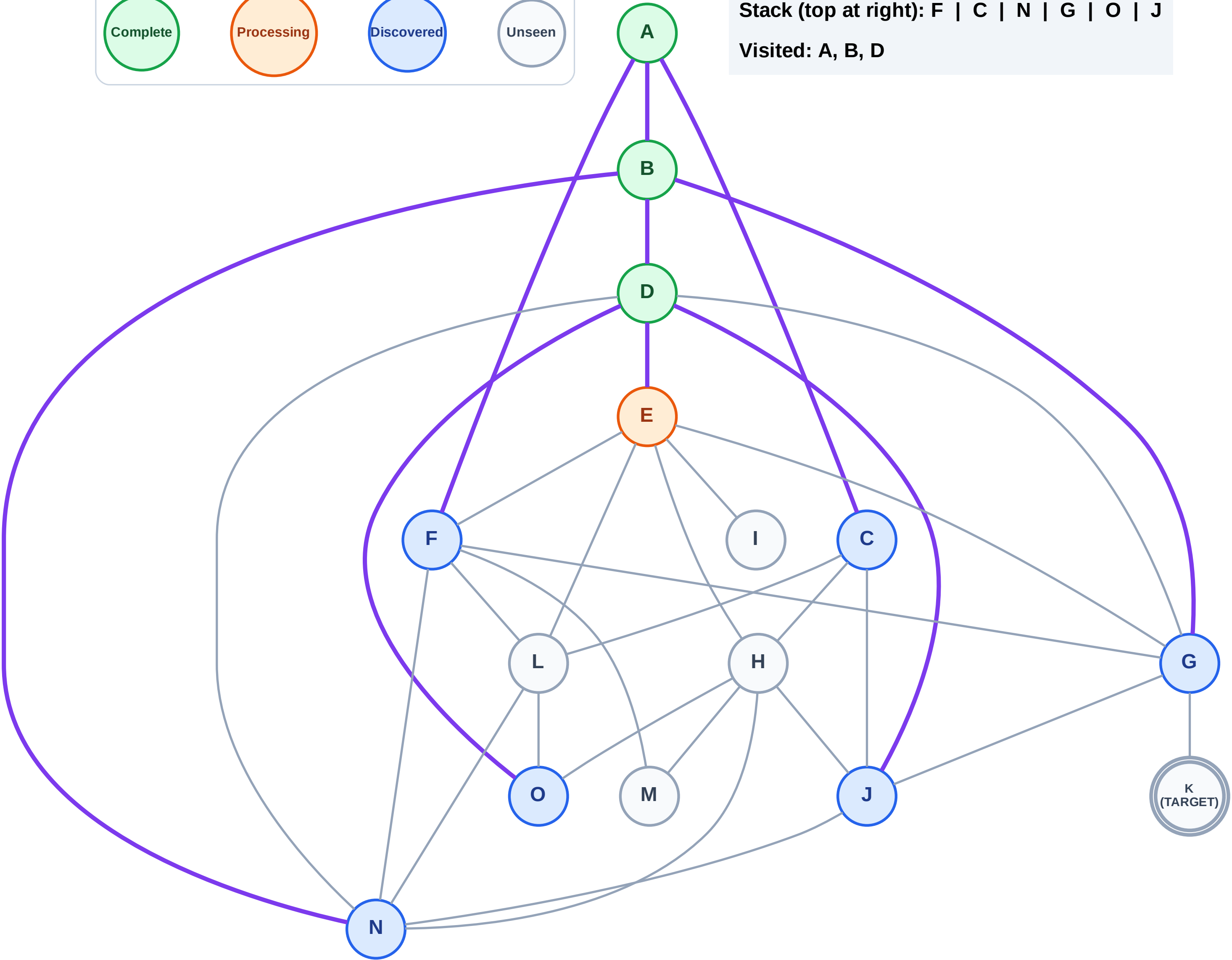
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J

Visited: A, B, D



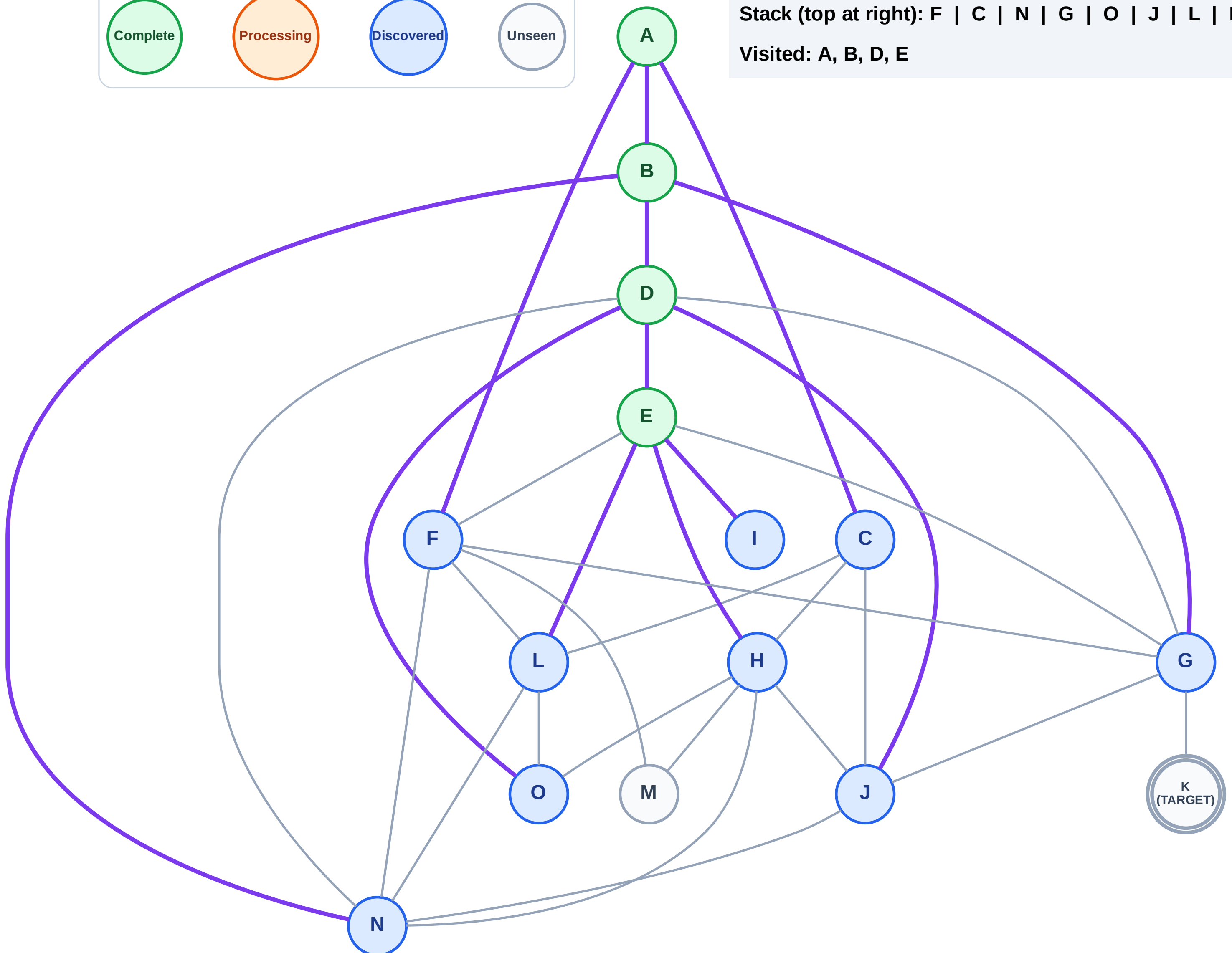
Step 9: Discover L, I, H from E

Stack (top at right): F | C | N | G | O | J | L | I | H

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.



DFS Traversal

Step 10: Process node H

Legend

Complete

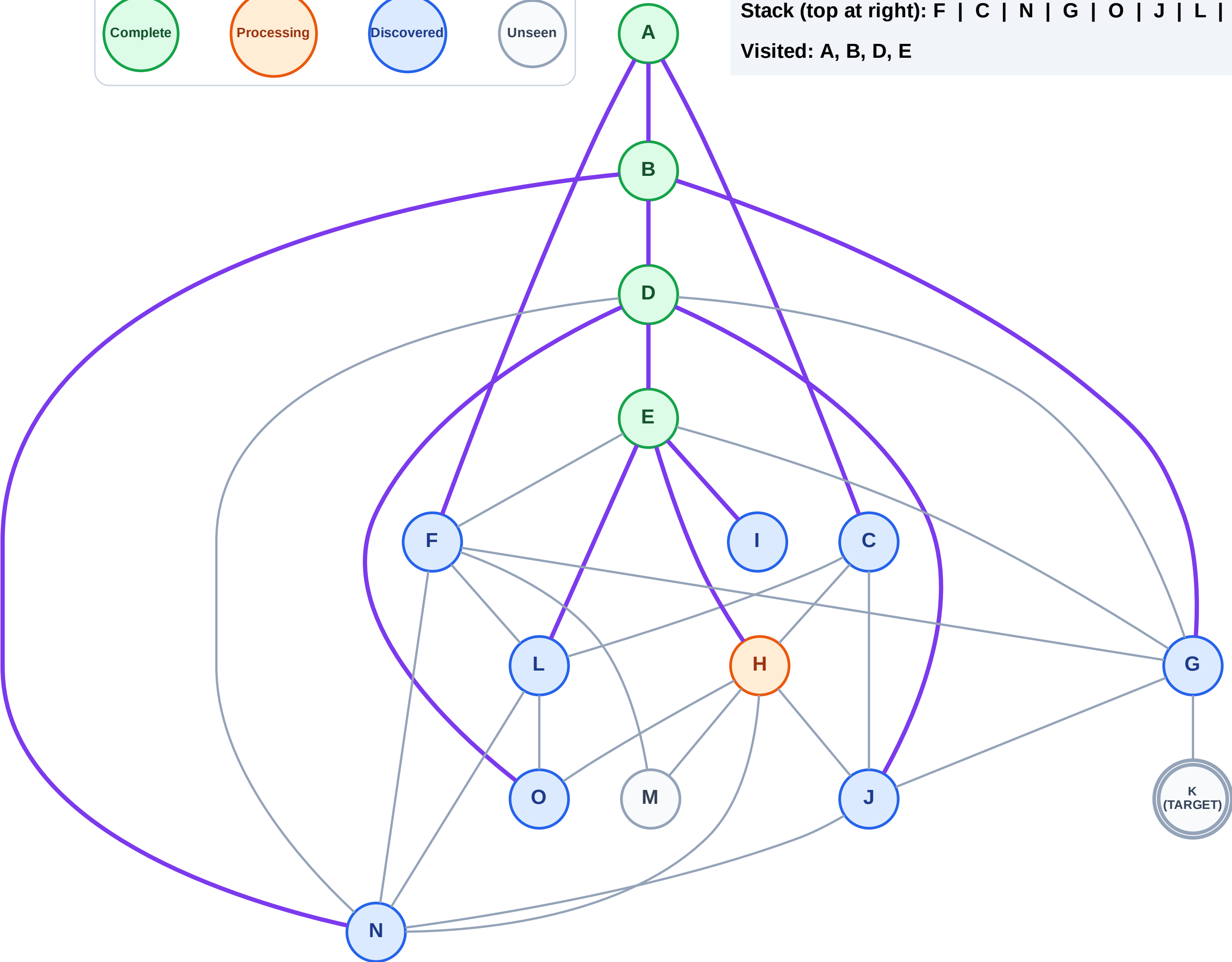
Processing

Discovered

Unseen

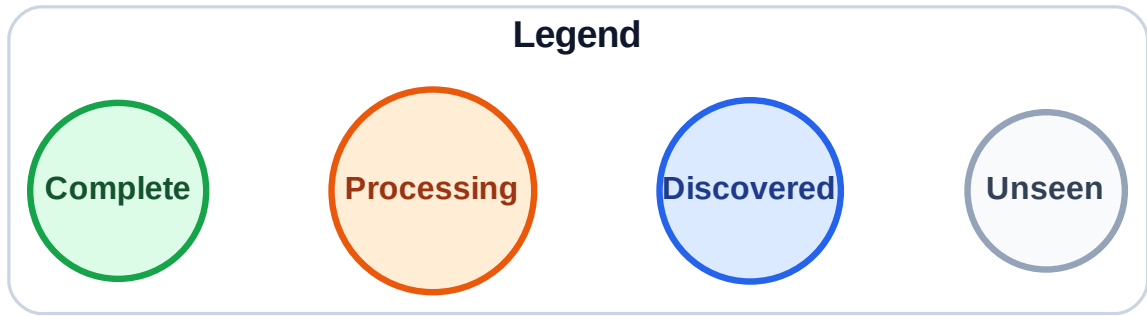
Stack (top at right): F | C | N | G | O | J | L | I

Visited: A, B, D, E



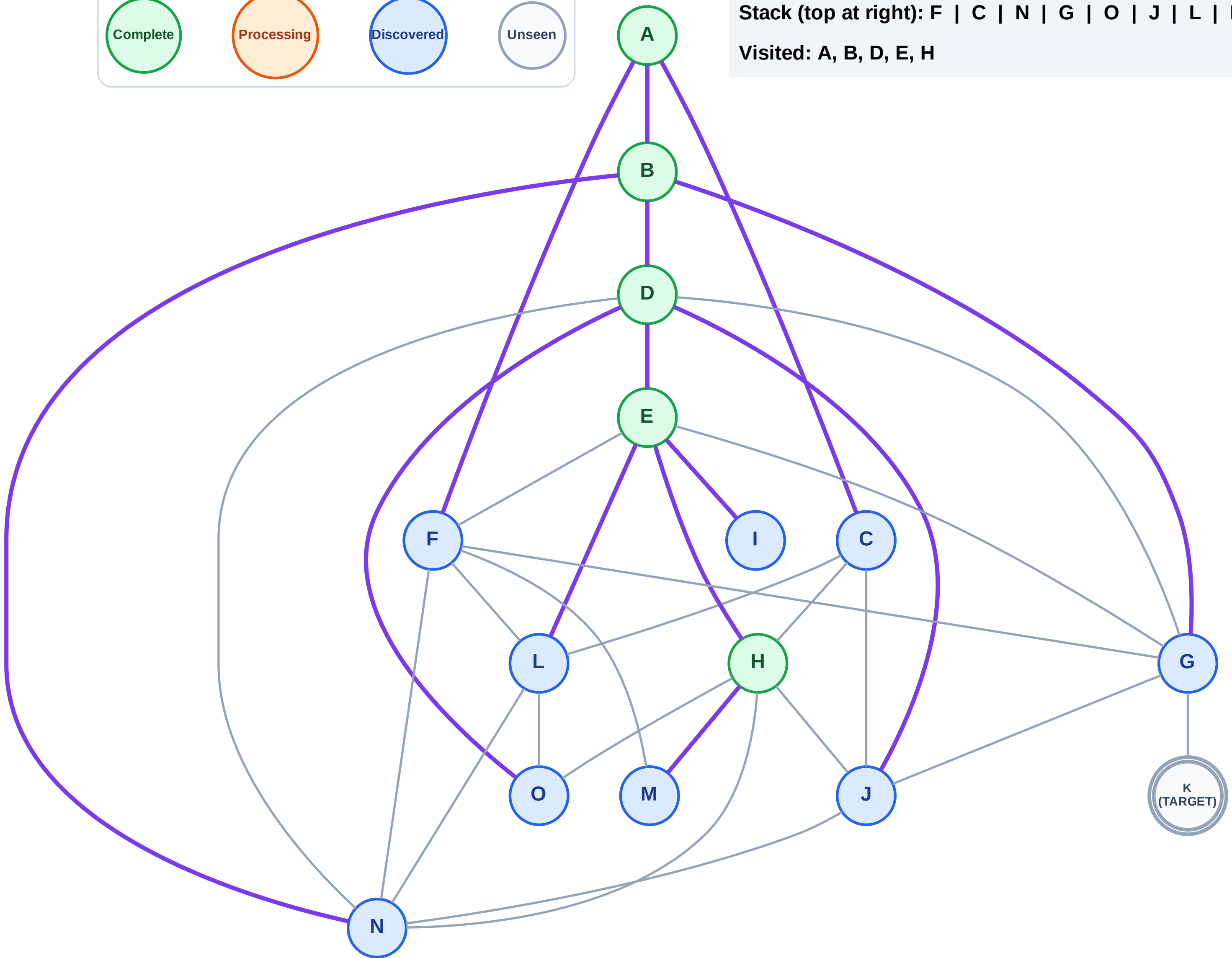
DFS Traversal

Step 11: Discover M from H

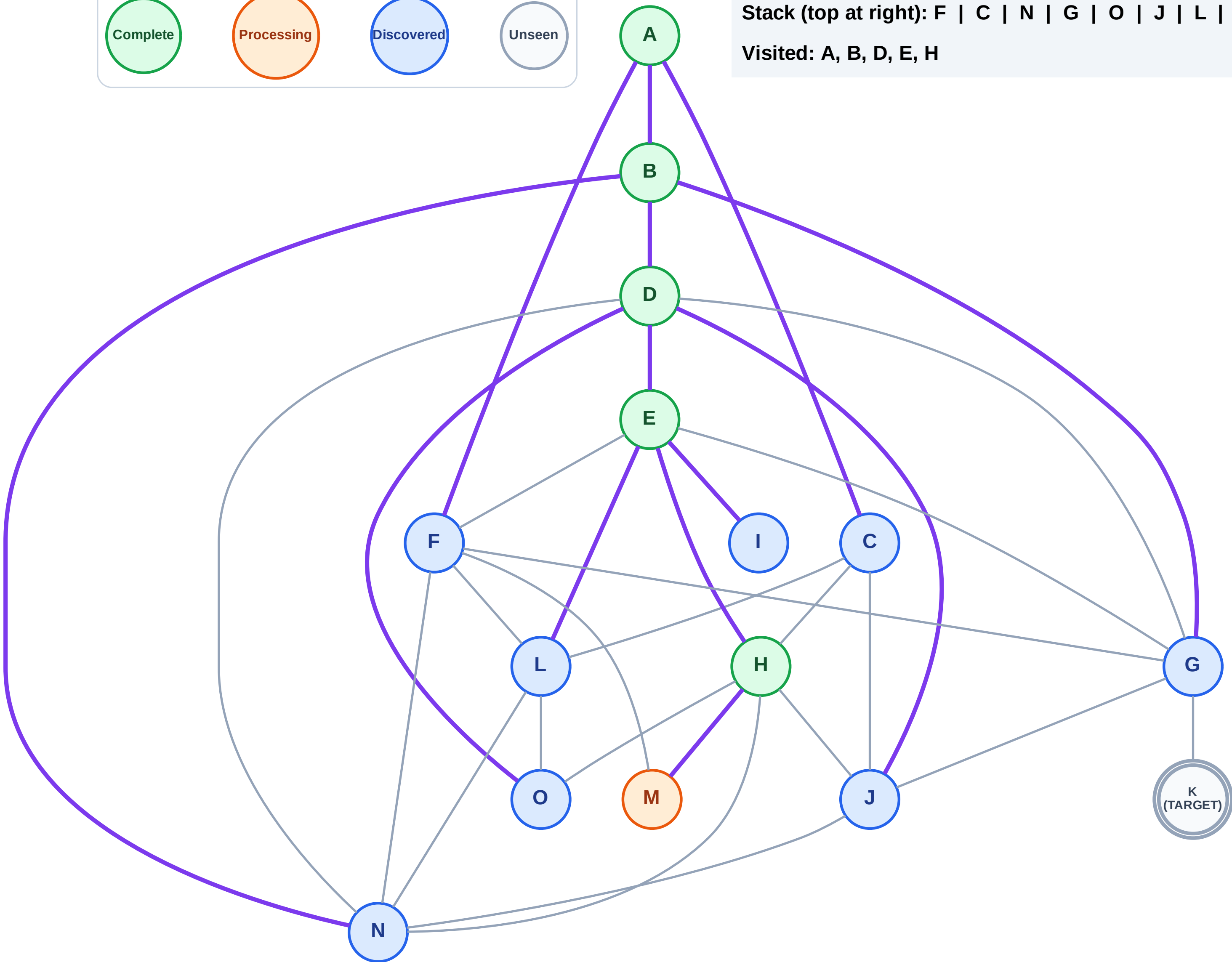


Stack (top at right): F | C | N | G | O | J | L | I | M

Visited: A, B, D, E, H



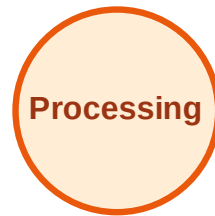
Step 12: Process node M



DFS Traversal

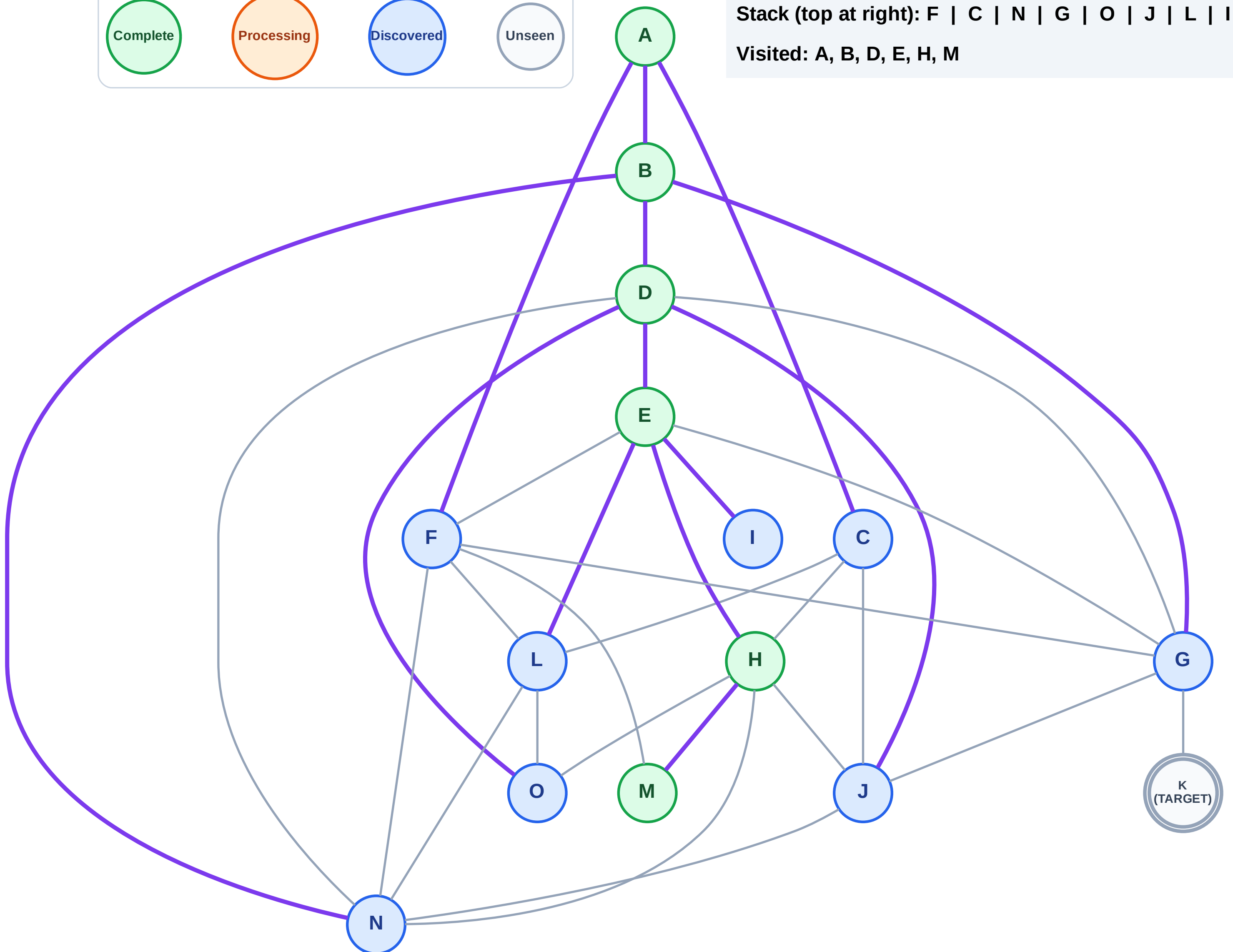
Step 13: No new neighbors from M

Legend



Stack (top at right): F | C | N | G | O | J | L | I

Visited: A, B, D, E, H, M



DFS Traversal

Step 14: Process node I

Legend

Complete

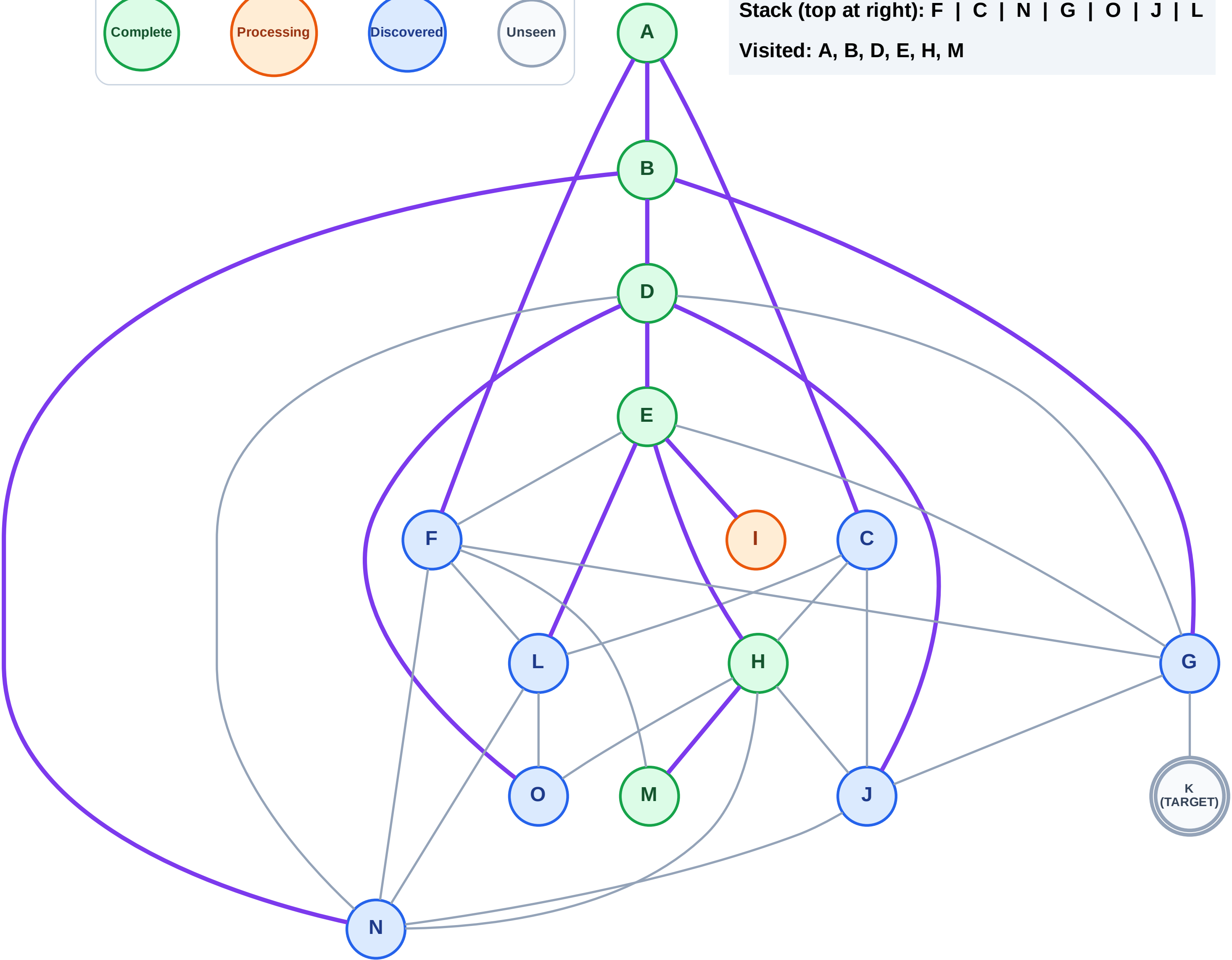
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J | L

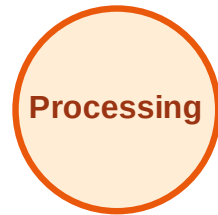
Visited: A, B, D, E, H, M



DFS Traversal

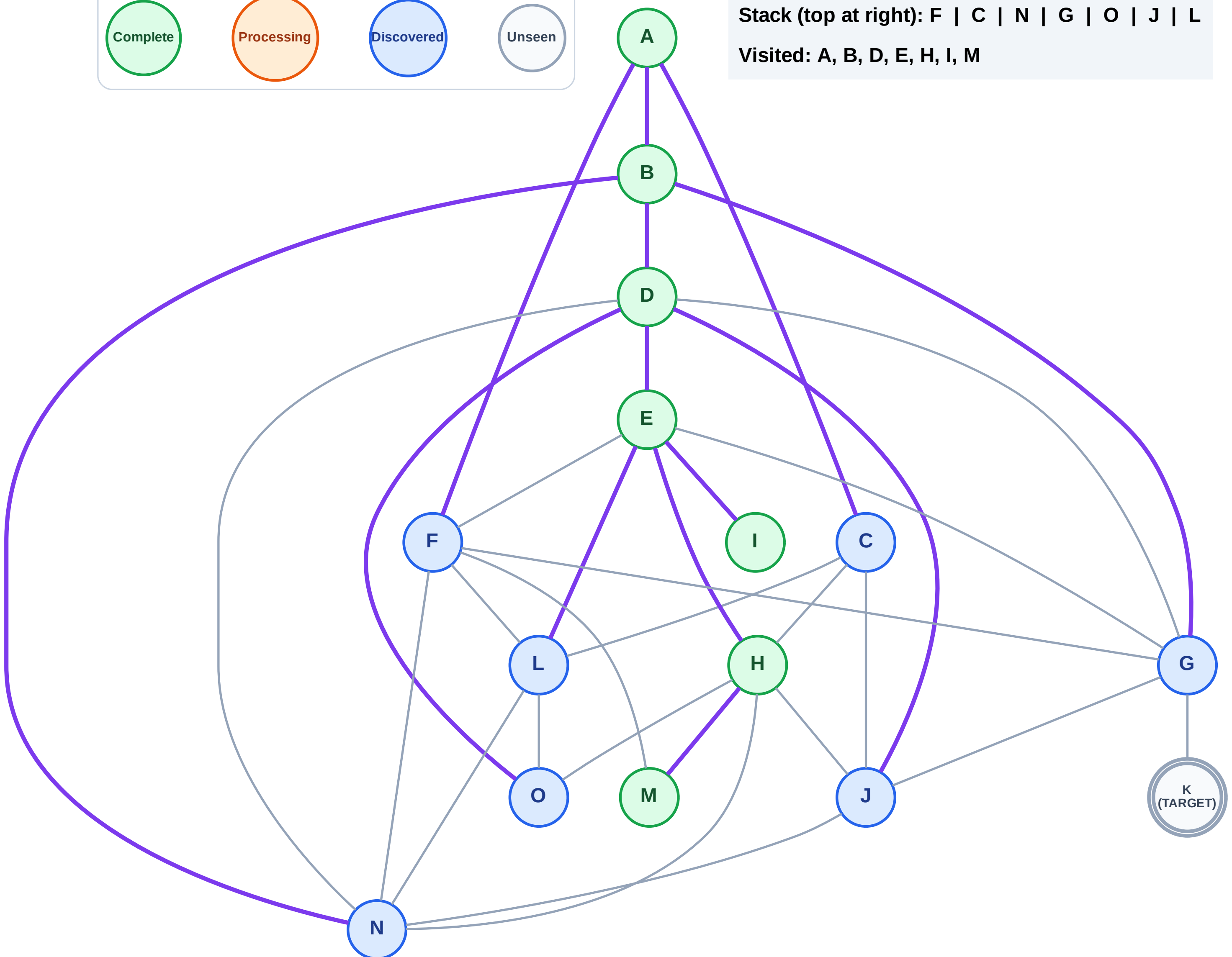
Step 15: No new neighbors from I

Legend



Stack (top at right): F | C | N | G | O | J | L

Visited: A, B, D, E, H, I, M



DFS Traversal

Step 16: Process node L

Legend

Complete

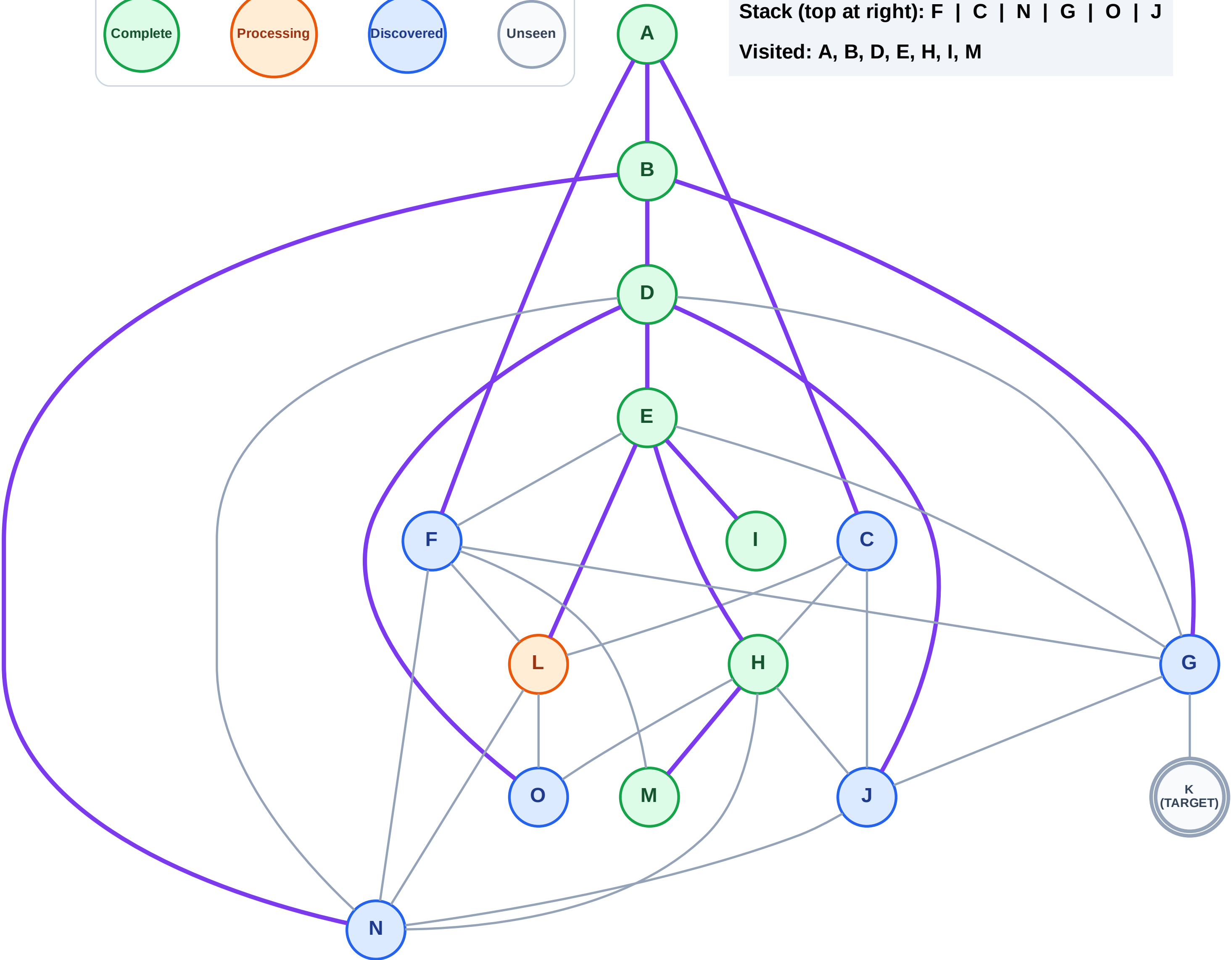
Processing

Discovered

Unseen

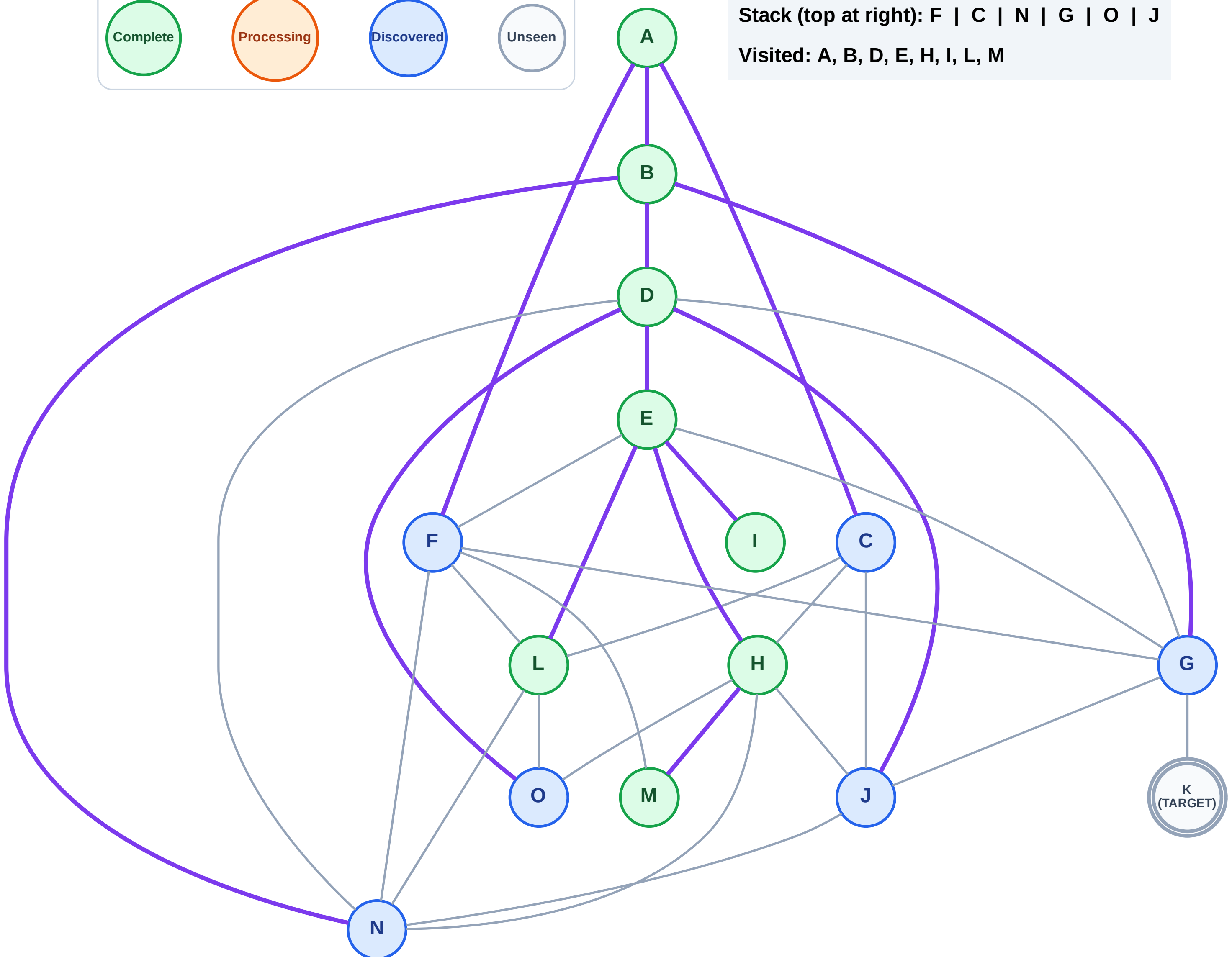
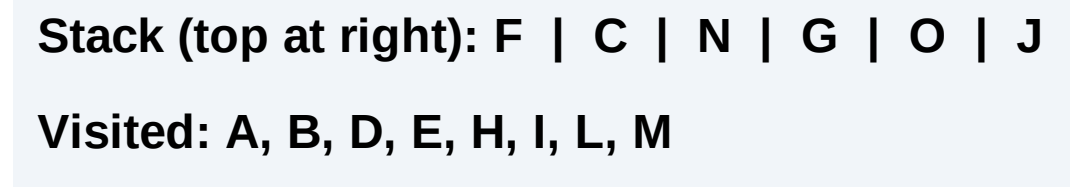
Stack (top at right): F | C | N | G | O | J

Visited: A, B, D, E, H, I, M



Step 17: No new neighbors from L

Step 17: No new neighbors from L



DFS Traversal

Step 18: Process node J

Legend

Complete

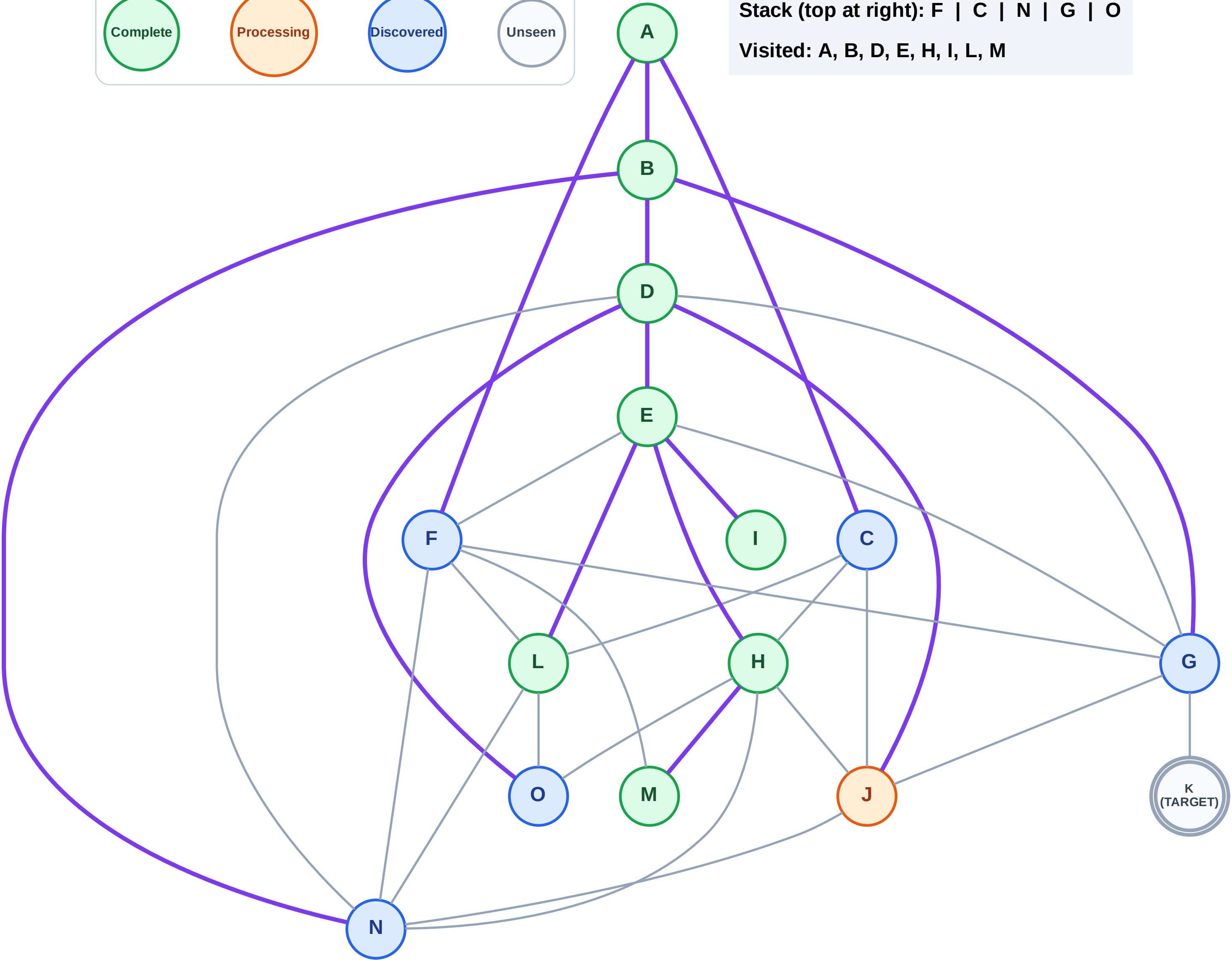
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O

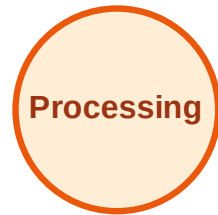
Visited: A, B, D, E, H, I, L, M



DFS Traversal

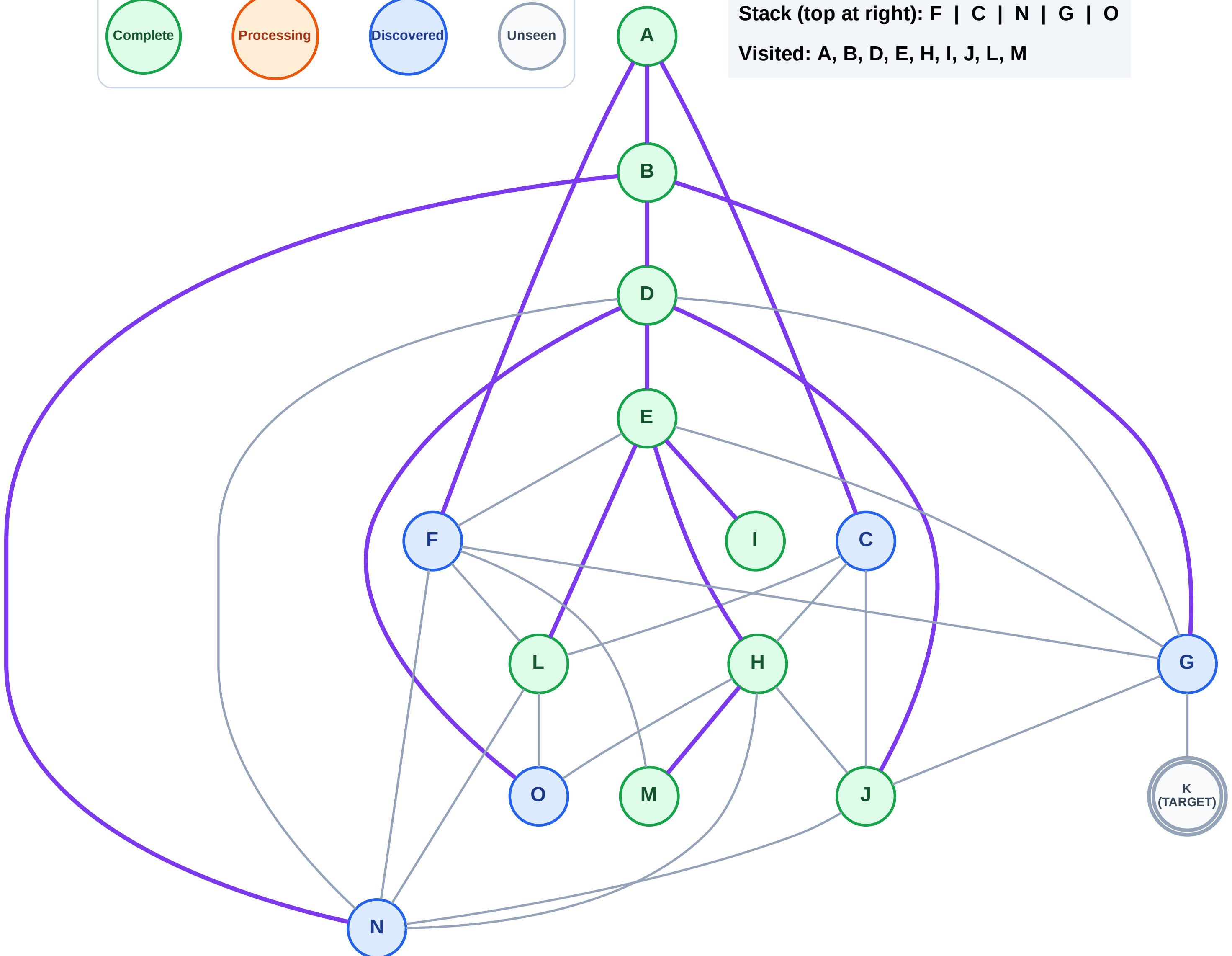
Step 19: No new neighbors from J

Legend



Stack (top at right): F | C | N | G | O

Visited: A, B, D, E, H, I, J, L, M



DFS Traversal

Step 20: Process node O

Legend

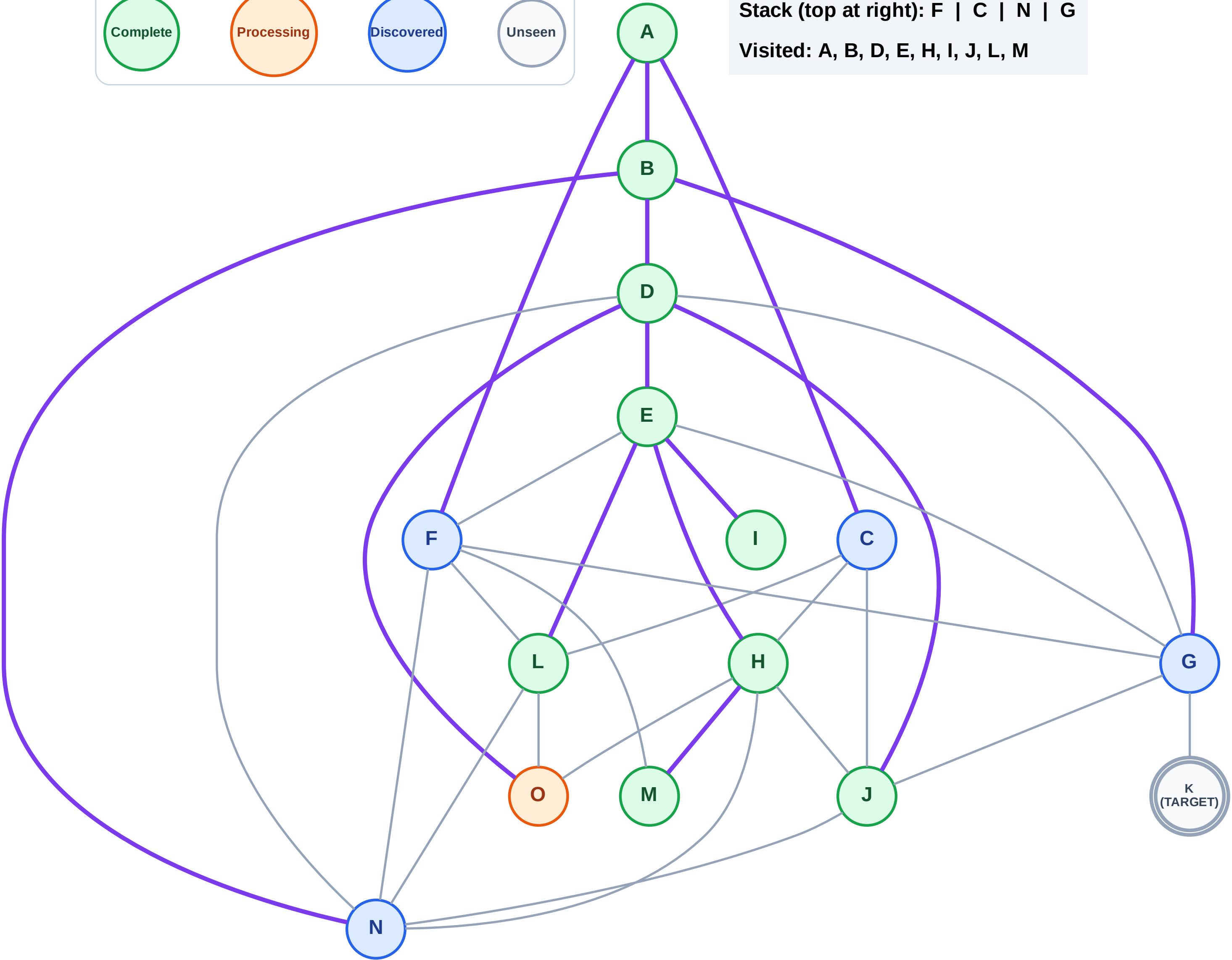
Complete

Processing

Discovered

Unseen

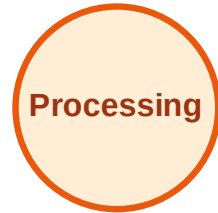
Stack (top at right): F | C | N | G
Visited: A, B, D, E, H, I, J, L, M



DFS Traversal

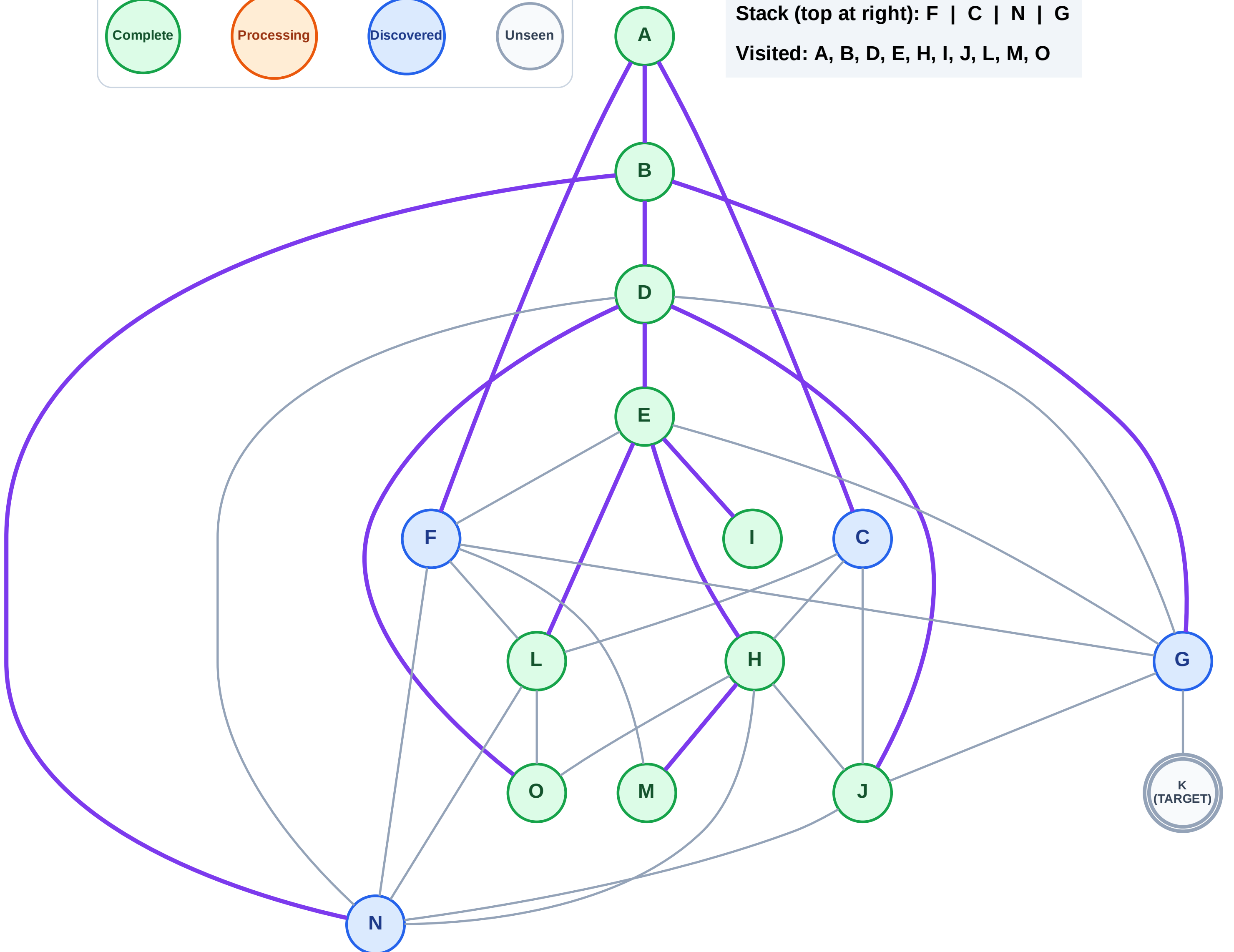
Step 21: No new neighbors from O

Legend



Stack (top at right): F | C | N | G

Visited: A, B, D, E, H, I, J, L, M, O



DFS Traversal

Step 22: Process node G

Legend

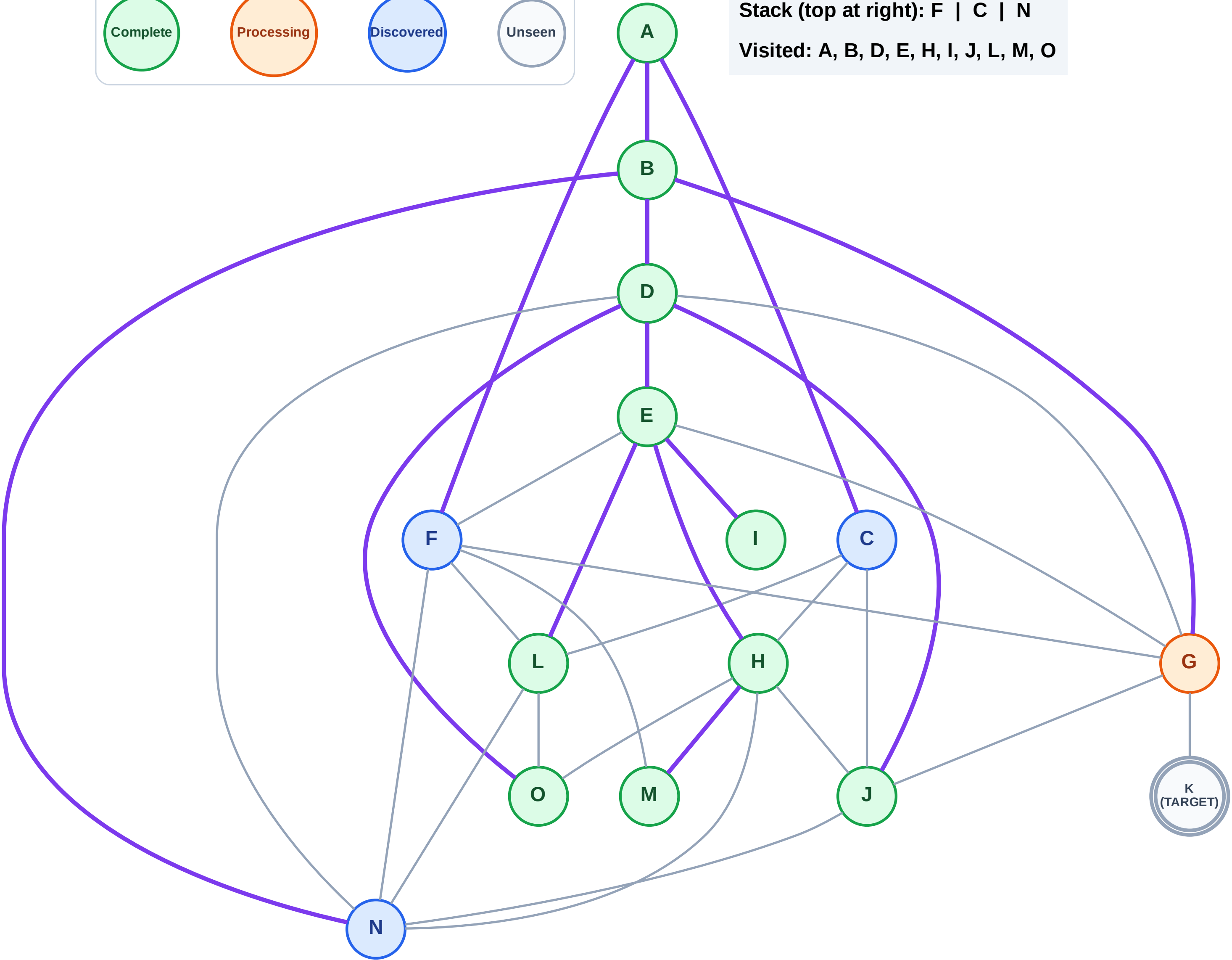
Complete

Processing

Discovered

Unseen

Stack (top at right): F | C | N
Visited: A, B, D, E, H, I, J, L, M, O



DFS Traversal

Step 23: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): F | C | N

Visited: A, B, D, E, G, H, I, J, K, L, M, O

